

Fourier Methods

Fourier Series

Recall: In 3D, the three basis vectors $\hat{x}, \hat{y}, \hat{z}$ are complete and orthonormal.

- Orthonormal $\rightarrow$ Orthogonal + Unit length

This means their inner products satisfy the orthonormality conditions

$$\hat{x} \cdot \hat{y} = \hat{y} \cdot \hat{z} = \hat{z} \cdot \hat{x} = 0$$

$$\hat{x} \cdot \hat{x} = \hat{y} \cdot \hat{y} = \hat{z} \cdot \hat{z} = 1$$

- Complete $\rightarrow$ Any vector V can be express as

$$V = a\hat{x} + b\hat{y} + c\hat{z}$$

By the orthonormality conditions, the coefficients a , b , and c can be obtained to be:

$$V \cdot \hat{x} = a\hat{x} \cdot \hat{x} + b\hat{y} \cdot \hat{x} + c\hat{z} \cdot \hat{x} = a$$

$$V \cdot \hat{y} = a\hat{x} \cdot \hat{y} + b\hat{y} \cdot \hat{y} + c\hat{z} \cdot \hat{y} = b$$

$$V \cdot \hat{z} = a\hat{x} \cdot \hat{z} + b\hat{y} \cdot \hat{z} + c\hat{z} \cdot \hat{z} = c$$

For real functions $g(x)$ and $h(x)$ in $-\pi \leq x \leq \pi$, it can be shown that

$$\langle g, h \rangle = \int_{-\pi}^{\pi} g(x)h(x)dx$$

is similarly a valid inner product.

Consider the real functions

- $\frac{1}{\sqrt{2\pi}}$
- $\frac{1}{\sqrt{\pi}} \cos x, \frac{1}{\sqrt{\pi}} \cos 2x, \frac{1}{\sqrt{\pi}} \cos 3x, \dots$
- $\frac{1}{\sqrt{\pi}} \sin x, \frac{1}{\sqrt{\pi}} \sin 2x, \frac{1}{\sqrt{\pi}} \sin 3x, \dots$

It can be easily shown that the functions are orthonormal.

Questions:

What criteria must an inner product satisfy?

Can you see the similarity between

$$\langle g, h \rangle = \int_{-\pi}^{\pi} g(x)h(x)dx$$

and

$$\vec{u} \cdot \vec{v} = u_x v_x + u_y v_y + u_z v_z?$$

Orthonormality relations:

$$\left\langle \frac{1}{\sqrt{2\pi}}, \frac{1}{\sqrt{2\pi}} \right\rangle = \int_{-\pi}^{\pi} \frac{1}{\sqrt{2\pi}} \times \frac{1}{\sqrt{2\pi}} dx = 1$$

$$\left\langle \frac{1}{\sqrt{\pi}} \cos nx, \frac{1}{\sqrt{\pi}} \cos nx \right\rangle = \int_{-\pi}^{\pi} \frac{1}{\sqrt{\pi}} \cos nx \times \frac{1}{\sqrt{\pi}} \cos nx dx = \frac{1}{\pi} \int_{-\pi}^{\pi} \cos^2 nx dx = 1$$

$$\left\langle \frac{1}{\sqrt{\pi}} \sin nx, \frac{1}{\sqrt{\pi}} \sin nx \right\rangle = \int_{-\pi}^{\pi} \frac{1}{\sqrt{\pi}} \sin nx \times \frac{1}{\sqrt{\pi}} \sin nx dx = \frac{1}{\pi} \int_{-\pi}^{\pi} \sin^2 nx dx = 1$$

$$\left\langle \frac{1}{\sqrt{2\pi}}, \frac{1}{\sqrt{\pi}} \cos nx \right\rangle = \int_{-\pi}^{\pi} \frac{1}{\sqrt{2\pi}} \times \frac{1}{\sqrt{\pi}} \cos nx dx = \frac{1}{\sqrt{2\pi}} \int_{-\pi}^{\pi} \cos nx dx = 0$$

$$\left\langle \frac{1}{\sqrt{2\pi}}, \frac{1}{\sqrt{\pi}} \sin nx \right\rangle = \int_{-\pi}^{\pi} \frac{1}{\sqrt{2\pi}} \times \frac{1}{\sqrt{\pi}} \sin nx dx = \frac{1}{\sqrt{2\pi}} \int_{-\pi}^{\pi} \sin nx dx = 0$$

$$\left\langle \frac{1}{\sqrt{\pi}} \cos nx, \frac{1}{\sqrt{\pi}} \cos mx \right\rangle = \int_{-\pi}^{\pi} \frac{1}{\sqrt{\pi}} \cos nx \times \frac{1}{\sqrt{\pi}} \cos mx dx = \frac{1}{\pi} \int_{-\pi}^{\pi} \cos nx \cos mx dx = 0 \quad \text{if } m \neq n$$

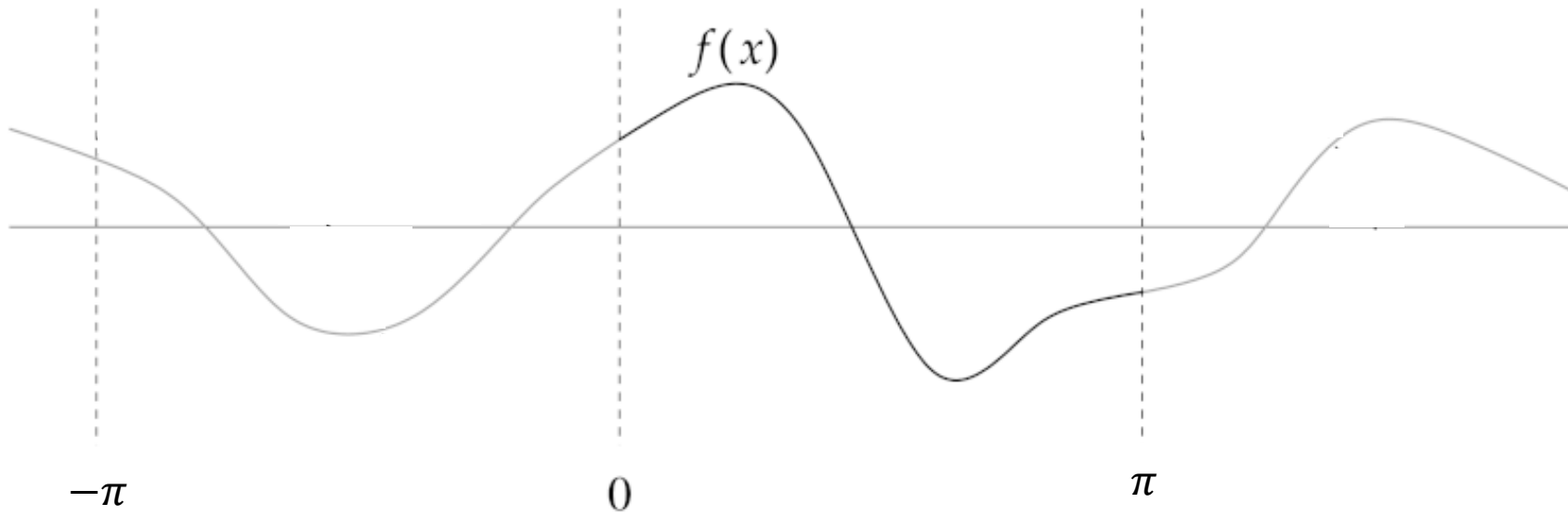
$$\left\langle \frac{1}{\sqrt{\pi}} \sin nx, \frac{1}{\sqrt{\pi}} \sin mx \right\rangle = \int_{-\pi}^{\pi} \frac{1}{\sqrt{\pi}} \sin nx \times \frac{1}{\sqrt{\pi}} \sin mx dx = \frac{1}{\pi} \int_{-\pi}^{\pi} \sin nx \sin mx dx = 0 \quad \text{if } m \neq n$$

$$\left\langle \frac{1}{\sqrt{\pi}} \cos nx, \frac{1}{\sqrt{\pi}} \sin mx \right\rangle = \int_{-\pi}^{\pi} \frac{1}{\sqrt{\pi}} \cos nx \times \frac{1}{\sqrt{\pi}} \sin mx dx = \frac{1}{\pi} \int_{-\pi}^{\pi} \cos nx \sin mx dx = 0 \quad \text{for all } m, n$$

It can be shown that these orthonormal also form a complete basis for periodic functions with period 2π satisfying certain conditions.

In other words, any periodic function f with period 2π satisfying these conditions can be expressed as

$$f(x) = c_0 \frac{1}{\sqrt{2\pi}} + c_1 \frac{1}{\sqrt{\pi}} \cos x + c_2 \frac{1}{\sqrt{\pi}} \cos 2x + c_3 \frac{1}{\sqrt{\pi}} \cos 3x + \dots \\ + d_1 \frac{1}{\sqrt{\pi}} \sin x + d_2 \frac{1}{\sqrt{\pi}} \sin 2x + d_3 \frac{1}{\sqrt{\pi}} \sin 3x + \dots$$



One can therefore use the orthonormality relations to obtain the coefficients:

$$\begin{aligned}c_0 &= \left\langle f(x), \frac{1}{\sqrt{2\pi}} \right\rangle = \int_{-\pi}^{\pi} \frac{1}{\sqrt{2\pi}} f(x) dx \\c_n &= \left\langle f(x), \frac{1}{\sqrt{\pi}} \cos n x \right\rangle = \int_{-\pi}^{\pi} \frac{1}{\sqrt{\pi}} f(x) \cos n x dx \\d_n &= \left\langle f(x), \frac{1}{\sqrt{\pi}} \sin n x \right\rangle = \int_{-\pi}^{\pi} \frac{1}{\sqrt{\pi}} f(x) \sin n x dx\end{aligned}$$

Hence,

$$\begin{aligned}f(x) &= \left(\int_{-\pi}^{\pi} \frac{1}{\sqrt{2\pi}} f(x) dx \right) \frac{1}{\sqrt{2\pi}} + \sum_{n=1}^{\infty} \left(\int_{-\pi}^{\pi} \frac{1}{\sqrt{\pi}} f(x) \cos n x dx \right) \frac{1}{\sqrt{\pi}} \cos n x \\&\quad + \sum_{n=1}^{\infty} \left(\int_{-\pi}^{\pi} \frac{1}{\sqrt{\pi}} f(x) \sin n x dx \right) \frac{1}{\sqrt{\pi}} \sin n x \\&= \frac{1}{2} \left(\frac{1}{\pi} \int_{-\pi}^{\pi} f(x) dx \right) + \sum_{n=1}^{\infty} \left(\frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos n x dx \right) \cos n x \\&\quad + \sum_{n=1}^{\infty} \left(\frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \sin n x dx \right) \sin n x\end{aligned}$$

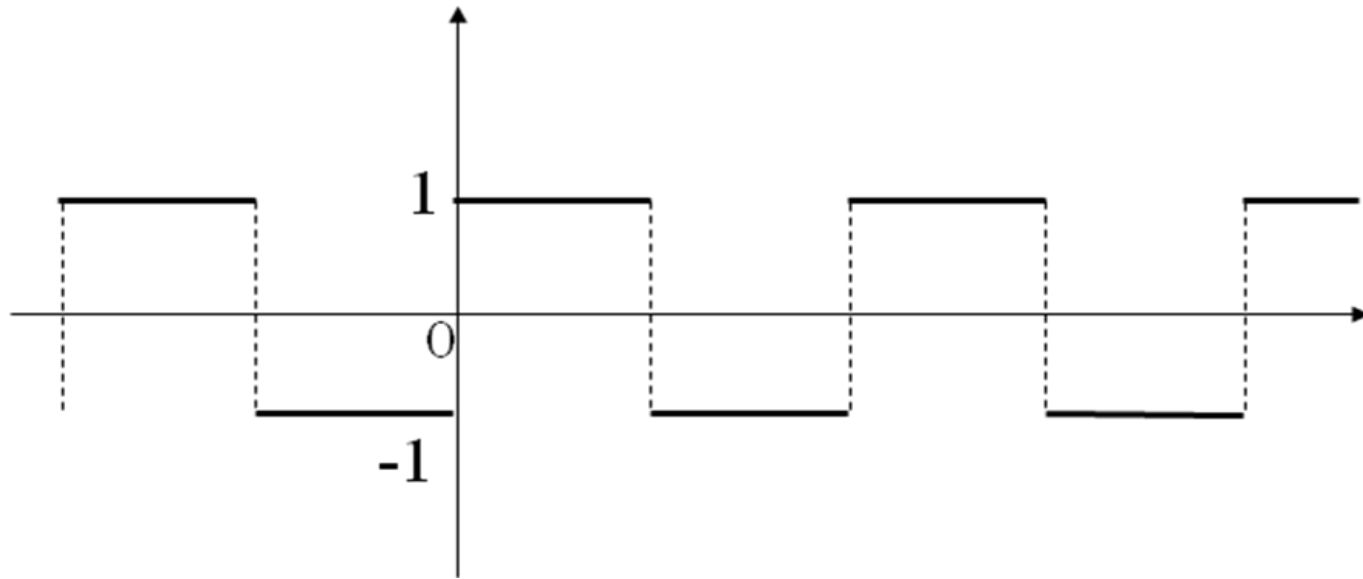
Example

Express the periodic function

$$f(x) = \begin{cases} V_0 & 0 < x < \pi \\ -V_0 & -\pi < x < 0 \end{cases}$$

$$f(x + 2\pi) = f(x) \quad \forall x$$

in terms of Fourier series.



Answer:

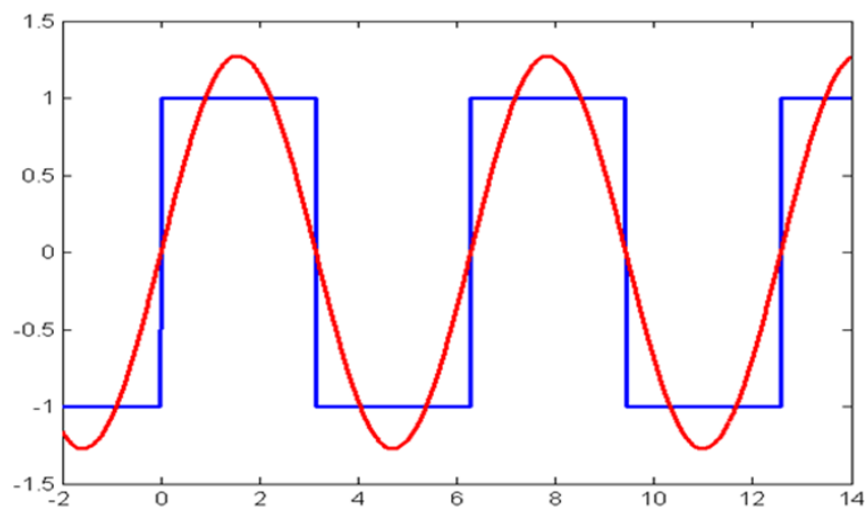
$$a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos n x dx = 0 \quad n = 0, 1, 2, 3, \dots$$
$$b_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \sin n x dx = \frac{2}{\pi} \int_0^{\pi} V_0 \sin n x dx = \frac{2V_0}{n\pi} (1 - \cos n \pi) \quad n = 1, 2, 3, \dots$$

- When n is even, $b_n = 0$.
- When n is odd, $b_n = \frac{4V_0}{n\pi} \quad n = 1, 3, 5, \dots$

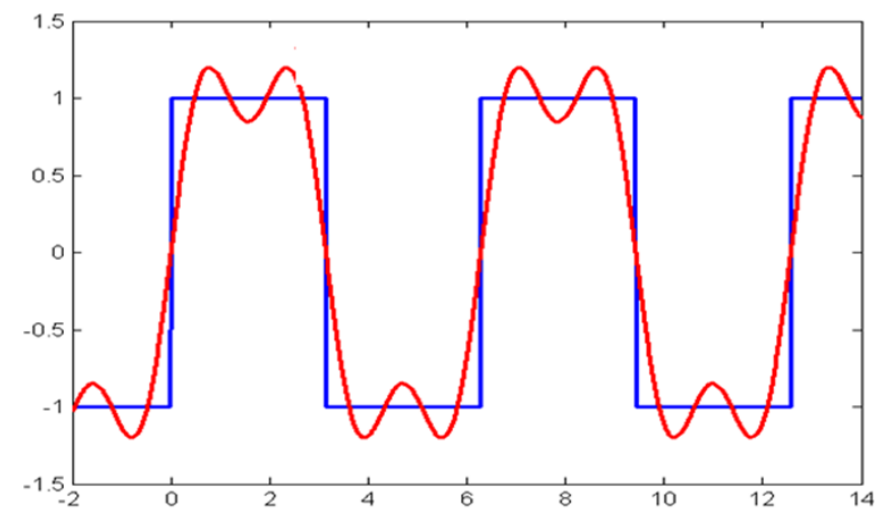
Hence

$$f(x) = \frac{4V_0}{\pi} \left(\sin x + \frac{1}{3} \sin 3x + \frac{1}{5} \sin 5x + \dots \right)$$

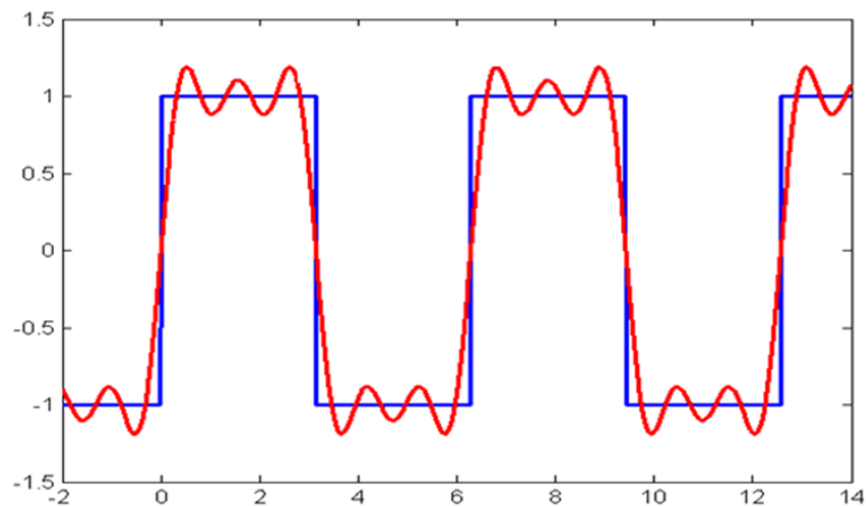
Using the first term of the series:



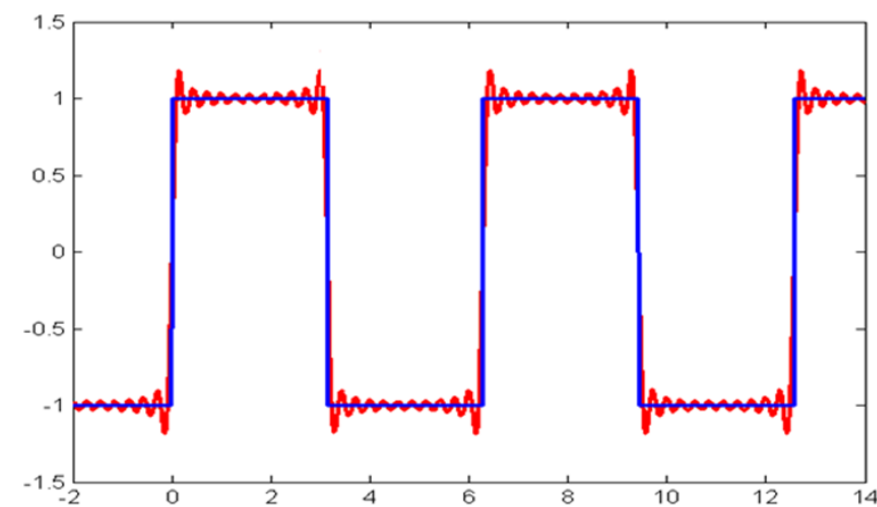
Using the first two terms of the series:



Using the first three terms of the series:



Using the first 12 terms of the series:



Other Intervals

For periodic function $f(x)$ with period $2l$ satisfying the same conditions stated in Dirichlet's theorem, let $y = \frac{\pi x}{l}$, then it becomes a periodic function in y with period 2π .

By the above theorem:

$$\begin{aligned} f(x) &= \frac{1}{2} a_0 + a_1 \cos y + a_2 \cos 2y + a_3 \cos 3y + \dots \\ &\quad + b_1 \sin y + b_2 \sin 2y + b_3 \sin 3y + \dots \\ &= \frac{1}{2} a_0 + a_1 \cos \frac{\pi x}{l} + a_2 \cos \frac{2\pi x}{l} + a_3 \cos \frac{3\pi x}{l} + \dots \\ &\quad + b_1 \sin \frac{\pi x}{l} + b_2 \sin \frac{2\pi x}{l} + b_3 \sin \frac{3\pi x}{l} + \dots \\ &= \frac{a_0}{2} + \sum_{n=1}^{\infty} \left(a_n \cos \frac{n\pi x}{l} + b_n \sin \frac{n\pi x}{l} \right) \end{aligned}$$

with

$$\begin{aligned} a_n &= \frac{1}{\pi} \int_{y=-\pi}^{y=\pi} f(x) \cos n y dy = \frac{1}{\pi} \int_{x=-l}^{x=l} f(x) \cos \frac{n\pi x}{l} d\left(\frac{\pi x}{l}\right) = \frac{1}{l} \int_{-l}^l f(x) \cos \frac{n\pi x}{l} dx & n = 0, 1, 2, 3, \dots \\ b_n &= \frac{1}{\pi} \int_{y=-\pi}^{y=\pi} f(x) \sin n y dy = \frac{1}{\pi} \int_{x=-l}^{x=l} f(x) \sin \frac{n\pi x}{l} d\left(\frac{\pi x}{l}\right) = \frac{1}{l} \int_{-l}^l f(x) \sin \frac{n\pi x}{l} dx & n = 1, 2, 3, \dots \end{aligned}$$

For periodic functions $f(x)$ with period $2l$

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} \left(a_n \cos \frac{n\pi x}{l} + b_n \sin \frac{n\pi x}{l} \right)$$

where

$$a_n = \frac{1}{l} \int_{-l}^l f(x) \cos \frac{n\pi x}{l} dx \quad n = 0, 1, 2, 3, \dots$$

$$b_n = \frac{1}{l} \int_{-l}^l f(x) \sin \frac{n\pi x}{l} dx \quad n = 1, 2, 3, \dots$$

Even Functions

In particular, if $f(x)$ is an even function, then

$$a_n = \frac{1}{l} \int_{-l}^l f(x) \cos \frac{n\pi x}{l} dx = \frac{2}{l} \int_0^l f(x) \cos \frac{n\pi x}{l} dx$$

$$b_n = \frac{1}{l} \int_{-l}^l f(x) \sin \frac{n\pi x}{l} dx = 0$$

For periodic even functions f with period $2l$:

$$f(x) = \frac{1}{2} a_0 + a_1 \cos \frac{\pi x}{l} + a_2 \cos \frac{2\pi x}{l} + a_3 \cos \frac{3\pi x}{l} + \dots$$

where

$$a_n = \frac{2}{l} \int_0^l f(x) \cos \frac{n\pi x}{l} dx \quad n = 0, 1, 2, 3, \dots$$

Odd Functions

On the other hand, if $f(x)$ is an odd function, then

$$a_n = \frac{1}{l} \int_{-l}^l f(x) \cos \frac{n\pi x}{l} dx = 0$$

$$b_n = \frac{1}{l} \int_{-l}^l f(x) \sin \frac{n\pi x}{l} dx = \frac{2}{l} \int_0^l f(x) \sin \frac{n\pi x}{l} dx$$

For periodic odd functions f with period $2l$:

$$f(x) = b_1 \sin \frac{\pi x}{l} + b_2 \sin \frac{2\pi x}{l} + b_3 \sin \frac{3\pi x}{l} + \dots$$

where

$$b_n = \frac{2}{l} \int_0^l f(x) \sin \frac{n\pi x}{l} dx \quad n = 1, 2, 3, \dots$$

Even and Odd Extension

One needs both sets of functions

$$\cos \frac{n\pi x}{l} \quad n = 0, 1, 2, 3, \dots$$

$$\sin \frac{n\pi x}{l} \quad n = 1, 2, 3, \dots$$

to expand arbitrary periodic functions with period $2l$ satisfying the Dirichlet's conditions.

For example, one cannot use the sine functions alone to expand $\cos \frac{\pi x}{l}$ in the interval $(-l, l)$, because it is orthogonal to all

$$\sin \frac{n\pi x}{l} \quad n = 1, 2, 3, \dots$$

inside this interval.

(Just like one cannot express $\hat{\mathbf{z}}$ in terms of $\hat{\mathbf{x}}$ and $\hat{\mathbf{y}}$, because it is orthogonal to both of them.)

In general, the sine functions $\sin \frac{n\pi x}{l}$ $n = 1, 2, 3, \dots$ alone are not complete in the interval $(-l, l)$.

(Just like $\hat{\mathbf{x}}$ and $\hat{\mathbf{y}}$ are not complete in the entire 3D space.)

However, they are “complete” in the half-interval $(0,l)$ in the sense that one can express any function $f(x)$ satisfying the Dirichlet’s conditions in $(0,l)$ with the sine functions $\sin \frac{n\pi x}{l}$ $n = 1,2,3, \dots$ alone.

(Just like $\hat{\mathbf{x}}$ and $\hat{\mathbf{y}}$ are complete in the subspace of the x - y plane.)

In fact, this subspace is the odd function with period $2l$ obtained by the odd extension of $f(x)$:

$$f_{\text{odd}}(x) = \begin{cases} f(x) & 0 < x \leq l \\ -f(-x) & -l \leq x < 0 \end{cases}$$

The odd function can then be expressed in terms of the sine series:

$$f_{\text{odd}}(x) = b_1 \sin \frac{\pi x}{l} + b_2 \sin \frac{2\pi x}{l} + b_3 \sin \frac{3\pi x}{l} + \dots$$

where

$$b_n = \frac{2}{l} \int_0^l f(x) \sin \frac{n\pi x}{l} dx \quad n = 1,2,3, \dots$$

Similarly, the cosine functions are “complete” in the half-interval $(0,l)$ in the sense that one can express any function $f(x)$ satisfying the Dirichlet’s conditions in $(0,l)$ with the cosine functions $\cos \frac{n\pi x}{l}$ $n = 0,1,2,3, \dots$ alone.

This time, this subspace is the even function with period $2l$ obtained by the even extension of $f(x)$:

$$f_{\text{even}}(x) = \begin{cases} f(x) & 0 < x \leq l \\ f(-x) & -l \leq x < 0 \end{cases}$$

The even function can then be expressed in terms of the cosine series:

$$f_{\text{even}}(x) = \frac{1}{2}a_0 + a_1 \cos \frac{\pi x}{l} + a_2 \cos \frac{2\pi x}{l} + a_3 \cos \frac{3\pi x}{l} + \dots$$

where

$$a_n = \frac{2}{l} \int_0^l f(x) \cos \frac{n\pi x}{l} dx \quad n = 0,1,2,3, \dots$$

Example

As mentioned, the functions $\sin nx$ can not expand $\cos x$ in $(-\pi, \pi)$. However, it can expand $\cos x$ in $(0, \pi)$ if we construct the odd-extension of $\cos x$ by defining

$$f(x) = \begin{cases} \cos x & 0 < x \leq \pi \\ -\cos x & -\pi \leq x < 0 \end{cases}$$
$$f(x + 2\pi) = f(x) \quad \forall x$$

which is an odd function with period 2π .

To find the Fourier coefficients, notice that when f is an odd function,

$$a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos nx dx = 0$$
$$b_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \sin nx dx = \frac{2}{\pi} \int_0^{\pi} f(x) \sin nx dx$$

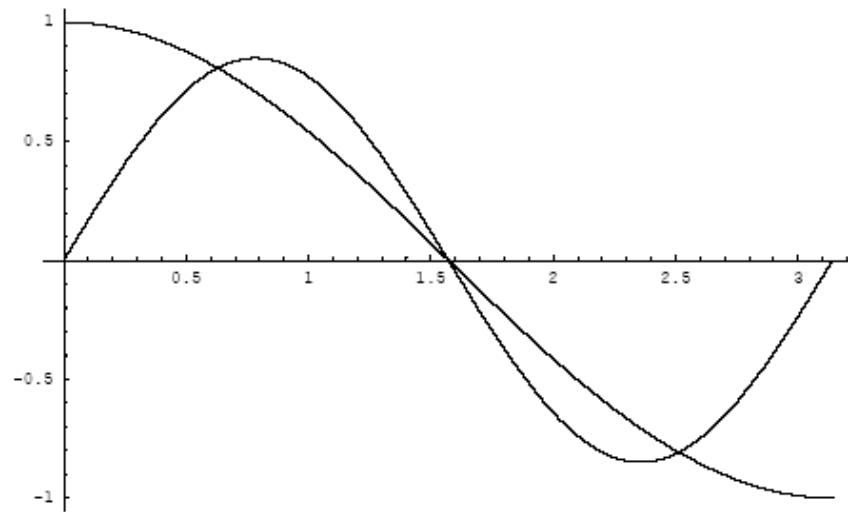
In particular, for the above function, we have

$$b_n = \frac{2}{\pi} \int_0^{\pi} \cos x \sin n x dx = \begin{cases} 0 & \text{if } n \text{ is odd} \\ \frac{4n}{(n^2 - 1)\pi} & \text{if } n \text{ is even} \end{cases}$$

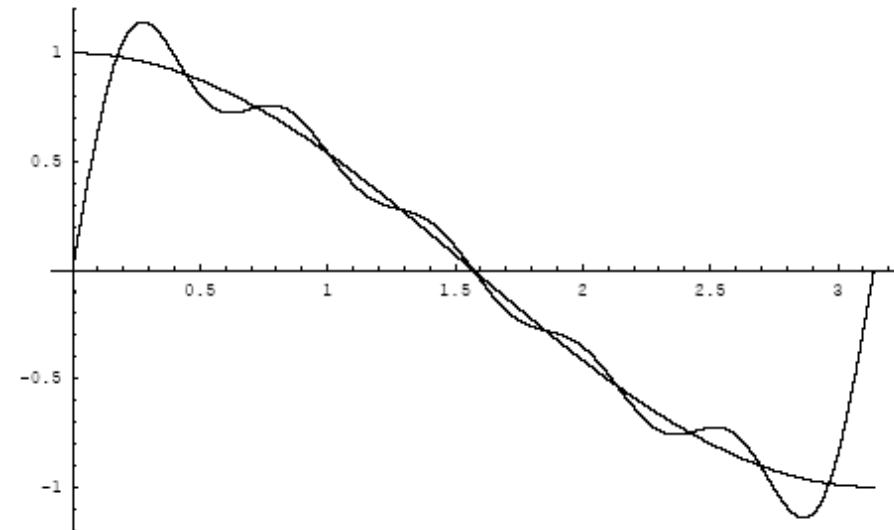
Therefore,

$$\cos x = \frac{4}{\pi} \sum_{n=2,4,6,\dots} \frac{n}{n^2 - 1} \sin n x$$

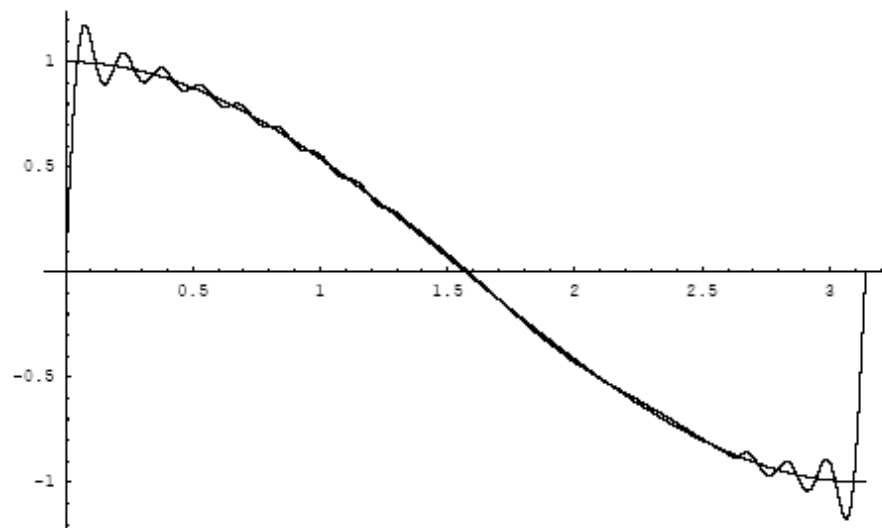
Using the first term of the series:



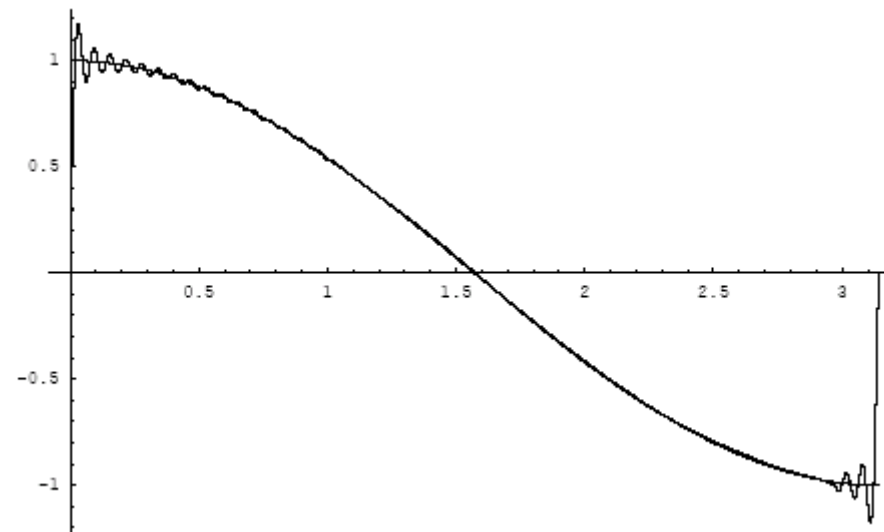
Using the first 5 terms of the series:



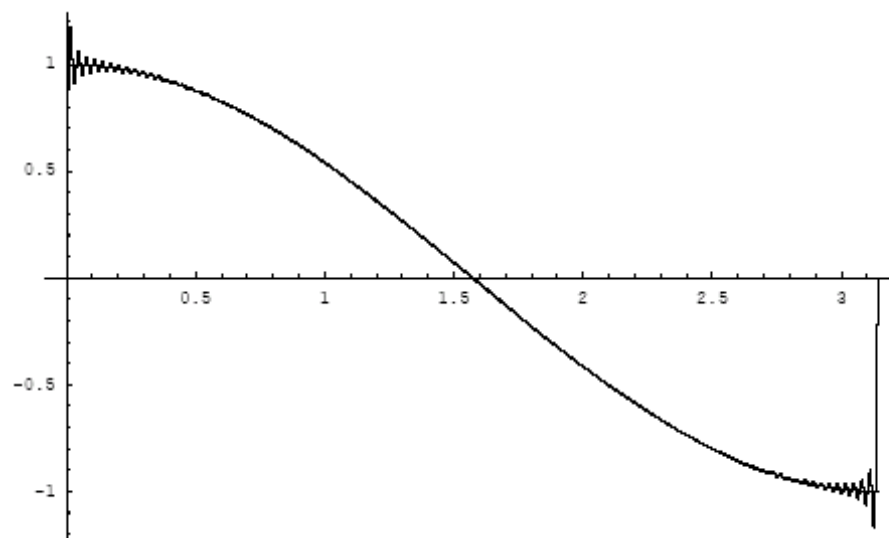
Using the first 20 term of the series:



Using the first 50 terms of the series:



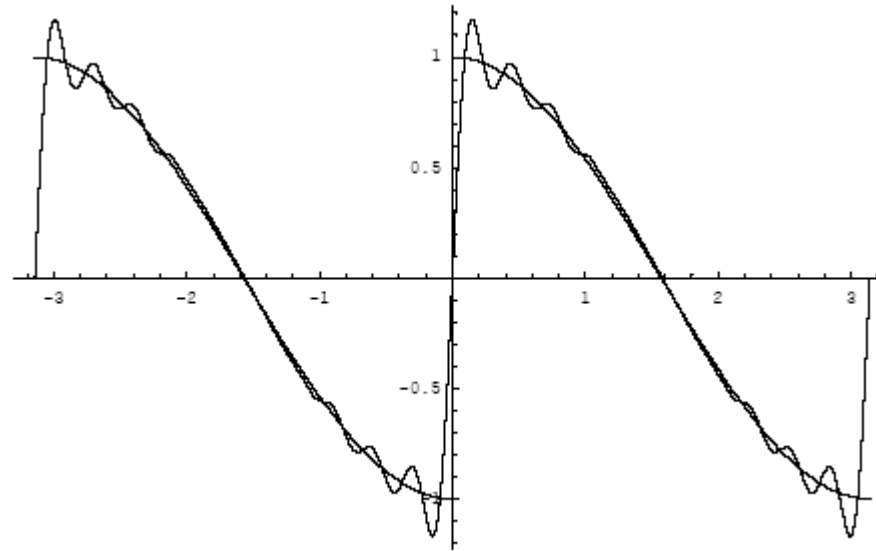
Using the first 100 terms of the series:



However, notice that the actual periodic function the series expands in $(-\pi, \pi)$ is not $\cos x$, but

$$f(x) = \begin{cases} \cos x & 0 \leq x \leq \pi \\ -\cos x & -\pi \leq x < 0 \end{cases}$$

If we plot the Fourier series in $(-\pi, \pi)$, we have



The Fourier series converges to the mid point at discontinuities of the original function, as stated by the theorem of Dirichlet.

Fourier Series

- We will now generalize the above real Fourier Series to complex functions

For complex functions $g(t)$ and $h(t)$ both with period T , it can be shown that

$$\langle g, h \rangle = \frac{1}{T} \int_{t_0}^{t_0+T} g(t) \overline{h(t)} dt$$

is a valid inner product.

(The factor $1/T$ is introduced here just for the sake of simplicity of the formulae to be derived)

Consider the complex functions $\phi_n(t) = e^{-in\omega_0 t}$

where $n = 0, \pm 1, \pm 2, \dots$ and $\omega_0 = \frac{2\pi}{T}$ so that they are all periodic functions with period T

We have

$$\langle \phi_n(t), \phi_n(t) \rangle = \frac{1}{T} \int_{t_0}^{t_0+T} e^{-in\omega_0 t} \overline{e^{-in\omega_0 t}} dt = \frac{1}{T} \int_{t_0}^{t_0+T} dt = 1$$

and for $n \neq m$,

$$\begin{aligned} \langle \phi_m(t), \phi_n(t) \rangle &= \frac{1}{T} \int_{t_0}^{t_0+T} e^{-im\omega_0 t} \overline{e^{-in\omega_0 t}} dt = \frac{1}{T} \int_{t_0}^{t_0+T} e^{i(n-m)\omega_0 t} dt \\ &= \frac{1}{T} \frac{1}{i(n-m)\omega_0} \left[e^{i(n-m)\omega_0 t} \right]_{t_0}^{t_0+T} = \frac{e^{i(n-m)\omega_0 t_0}}{2\pi i(n-m)} (e^{i(n-m)\omega_0 T} - 1) \\ &= \frac{e^{i(n-m)\omega_0 t_0}}{2\pi i(n-m)} (e^{2\pi i(n-m)} - 1) \\ &= 0 \end{aligned}$$

i.e.,

$$\langle \phi_m(t), \phi_n(t) \rangle = \delta_{mn}$$

Fourier Series

It can be shown that $\phi_n(t)$ form a complete basis for complex T -periodic functions $h(t)$ satisfying similar Dirichlet conditions.

That is, these functions admit Fourier series expansions and can be expressed as

$$h(t) = \sum_{n=-\infty}^{\infty} H_n e^{-in\omega_0 t}$$

- ω_0 : fundamental frequency
- H_n : Fourier coefficients

- The series is interpreted as $h(t) = \lim_{N \rightarrow \infty} \sum_{n=-N}^N H_n e^{-in\omega_0 t}$

Fourier Coefficients

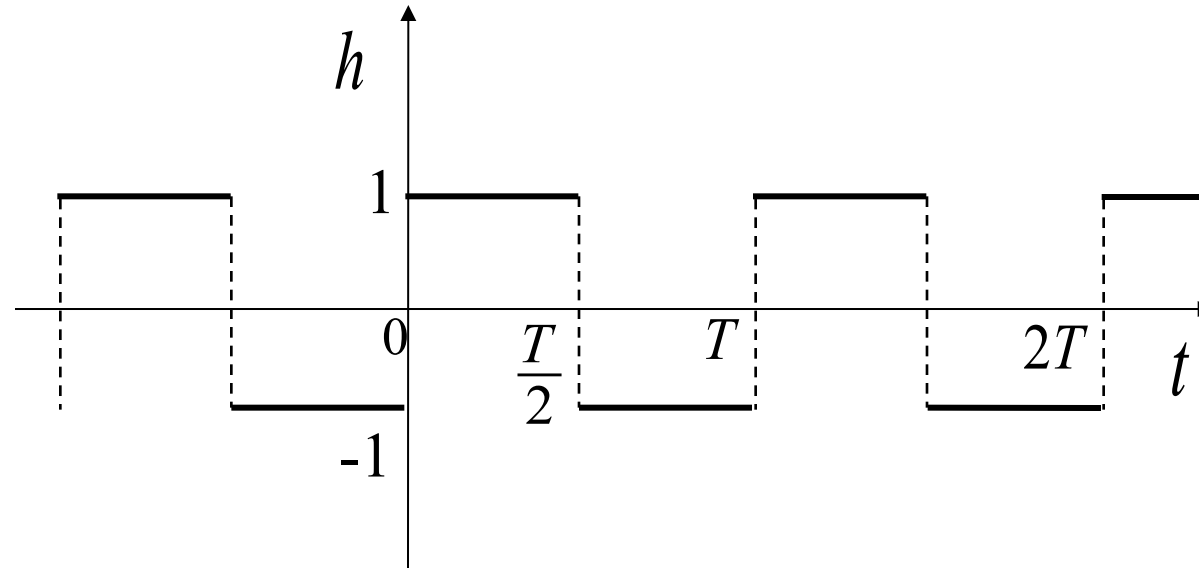
$$\langle h(t), \phi_n(t) \rangle = \left\langle \sum_{m=-\infty}^{\infty} H_m e^{-im\omega_0 t}, e^{-in\omega_0 t} \right\rangle$$

$$\frac{1}{T} \int_{t_0}^{t_0+T} h(t) \overline{e^{-in\omega_0 t}} dt = \sum_{m=-\infty}^{\infty} H_m \langle e^{-im\omega_0 t}, e^{-in\omega_0 t} \rangle$$

$$\frac{1}{T} \int_{t_0}^{t_0+T} h(t) e^{in\omega_0 t} dt = \sum_{m=-\infty}^{\infty} H_m \delta_{mn}$$

$$H_n = \frac{1}{T} \int_{t_0}^{t_0+T} h(t) e^{in\omega_0 t} dt$$

Example: Square Wave:



$$h(t) = \sum_{\substack{n=-\infty \\ n \text{ odd}}}^{\infty} \frac{2i}{n\pi} e^{-in\omega_0 t}$$



Derive this

Define
$$h_N(t) = \sum_{\substack{n=-N \\ n \text{ odd}}}^N \frac{2i}{n\pi} e^{-in\omega_0 t},$$

then
$$h(t) = \lim_{N \rightarrow \infty} h_N(t)$$

Fourier Series - Summary

Given a function $h(t)$, of period T , fundamental frequency $\omega_0 = \frac{2\pi}{T}$. We find the Fourier expansion by computing

$$H_n = \frac{1}{T} \int_{t_0}^{t_0+T} h(t) e^{in\omega_0 t} dt$$

$$h(t) = \sum_{n=-\infty}^{\infty} H_n e^{-in\omega_0 t}$$

In short: all “physical” periodic functions are superpositions of sinusoids, and the above formula gives the way to find the right superposition for a given function.

Representation of the Fourier Coefficients

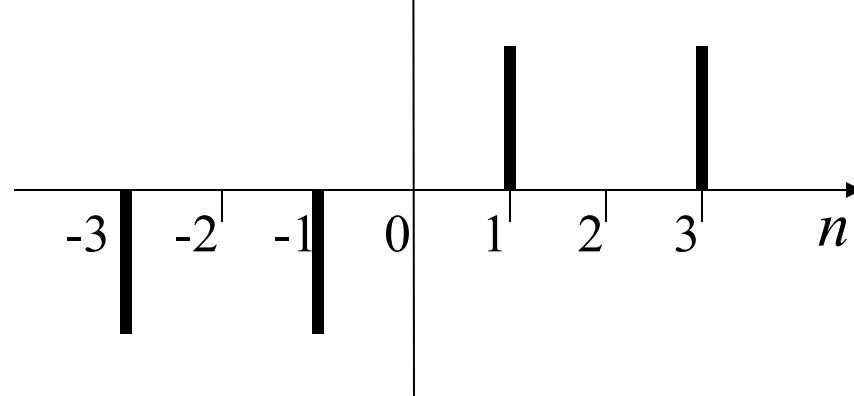
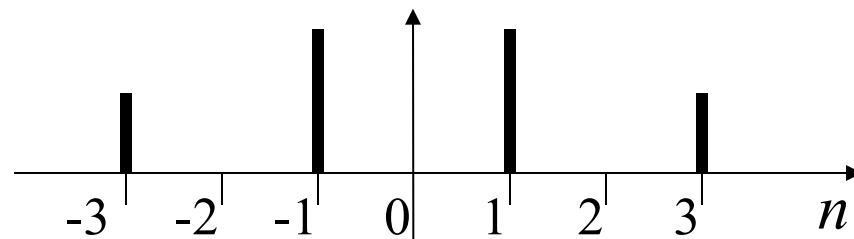
H_n is a complex number, $H_n = |H_n| e^{i\theta_n}$

The amplitude and phase (line) spectra are plots of $|H_n|$ and θ_n as a function of n .

For the square wave example: $H_n = \begin{cases} 0 & \text{if } n \text{ is even.} \\ \frac{2i}{n\pi} & \text{if } n \text{ is odd.} \end{cases}$

$$|H_n| = \begin{cases} 0 & \text{if } n \text{ is even.} \\ \frac{2}{|n|\pi} & \text{if } n \text{ is odd.} \end{cases}$$

$$\theta_n = \begin{cases} \text{undefined} & \text{if } n \text{ is even.} \\ \frac{\pi}{2} \text{sign}(n) & \text{if } n \text{ is odd.} \end{cases}$$



Symmetries in Line Spectra

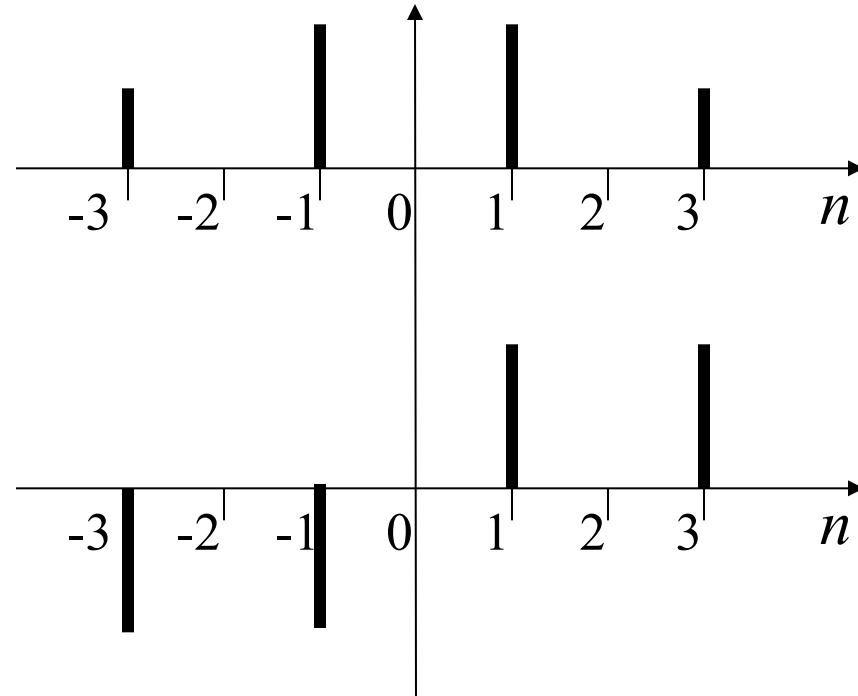
If $h(t)$ is real-valued, then $H_{-n} = \overline{H_n}$

$$H_{-n} = \frac{1}{T} \int_{t_0}^{t_0+T} h(t) e^{-in\omega_0 t} dt = \frac{1}{T} \int_{t_0}^{t_0+T} \overline{h(t) e^{in\omega_0 t}} dt = \overline{\frac{1}{T} \int_{t_0}^{t_0+T} h(t) e^{in\omega_0 t} dt} = \overline{H_n}$$

Consequence:

$|H_n| = |H_{-n}|$: the amplitude is an even function of n .

$\theta_n = -\theta_{-n}$: the phase is an odd function of n .



Fourier Series with Trigonometric Functions

If $h(t)$ is real-valued, the above reduces to the trigonometric Fourier Series derived earlier

$$\begin{aligned} H_n &= \frac{1}{T} \int_{t_0}^{t_0+T} h(t) e^{in\omega_0 t} dt = \frac{1}{T} \int_{t_0}^{t_0+T} h(t) [\cos(n\omega_0 t) + i \sin(n\omega_0 t)] dt \\ &= \underbrace{\frac{1}{T} \int_{t_0}^{t_0+T} h(t) \cos(n\omega_0 t) dt}_{a_n} + i \underbrace{\frac{1}{T} \int_{t_0}^{t_0+T} h(t) \sin(n\omega_0 t) dt}_{b_n} = a_n + ib_n \end{aligned}$$

In particular, for $n = 0$, we have $H_0 = a_0$

$$\begin{aligned} \text{Also, } h(t) &= \sum_{n=-\infty}^{\infty} H_n e^{-in\omega_0 t} = H_0 + \sum_{n=1}^{\infty} \left(H_n e^{-in\omega_0 t} + H_{-n} e^{in\omega_0 t} \right) \\ &= \underbrace{a_0}_{H_{-n}=\overline{H_n}} + \sum_{n=1}^{\infty} \left(H_n e^{-in\omega_0 t} + \overline{H_n e^{-in\omega_0 t}} \right) = a_0 + 2 \sum_{n=1}^{\infty} \operatorname{Re} \left[H_n e^{-in\omega_0 t} \right] \end{aligned}$$

Now,

$$\begin{aligned}\operatorname{Re}\left[H_n e^{-in\omega_0 t}\right] &= \operatorname{Re}\left[(a_n + ib_n)[\cos(n\omega_0 t) - i \sin(n\omega_0 t)]\right] \\ &= a_n \cos(n\omega_0 t) + b_n \sin(n\omega_0 t)\end{aligned}$$

Therefore
$$h(t) = a_0 + 2 \sum_{n=1}^{\infty} (a_n \cos(n\omega_0 t) + b_n \sin(n\omega_0 t)),$$

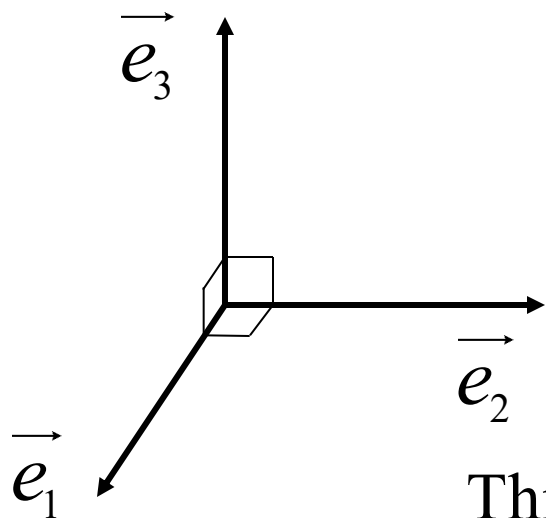
where
$$a_n = \frac{1}{T} \int_{t_0}^{t_0+T} h(t) \cos(n\omega_0 t) dt, \quad b_n = \frac{1}{T} \int_{t_0}^{t_0+T} h(t) \sin(n\omega_0 t) dt.$$

Fourier series with trigonometric functions, for real $h(t)$

The term $a_n \cos(n\omega_0 t) + b_n \sin(n\omega_0 t)$ in the sum is called the " n -th harmonic" of the function $h(t)$.

Interpretation of Fourier Series: Expansion in an Orthonormal Basis

Analogy: orthonormal basis in 3-dimensional space:



Vectors $\vec{e}_1, \vec{e}_2, \vec{e}_3$ are orthonormal when they:

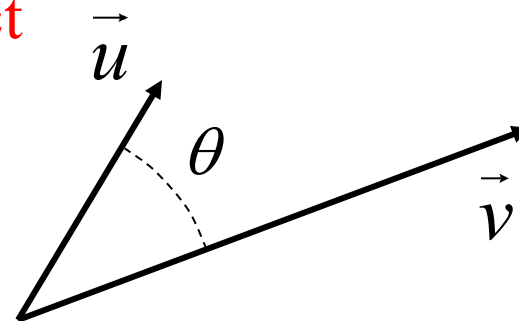
- are mutually orthogonal: $\vec{e}_m \perp \vec{e}_n$ for $m \neq n$
- have unit length: $|\vec{e}_n| = 1$ for $n = 1, 2, 3$.

This can be expressed as $\langle \vec{e}_m, \vec{e}_n \rangle = \begin{cases} 0 & \text{if } m \neq n \\ 1 & \text{if } m = n \end{cases}$,

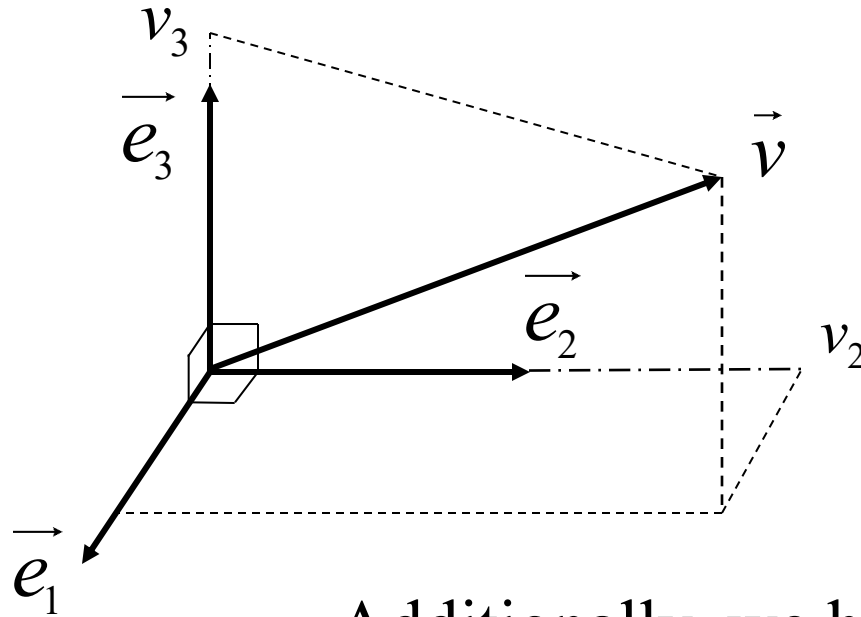
where $\langle \vec{u}, \vec{v} \rangle$ denotes the **inner product**

(or dot product) of vectors $\vec{u}, \vec{v}$.

Recall, $\langle \vec{u}, \vec{v} \rangle = |\vec{u}| |\vec{v}| \cos(\theta)$



Expanding a vector in an orthonormal basis



$$\vec{v} = v_1 \vec{e}_1 + v_2 \vec{e}_2 + v_3 \vec{e}_3,$$

where the scalar coefficients

v_1, v_2, v_3 can be expressed as

$$v_n = \langle \vec{v}, \vec{e}_n \rangle \text{ for } n = 1, 2, 3.$$

Additionally, we have the relationship

$$\langle \vec{u}, \vec{v} \rangle = u_1 v_1 + u_2 v_2 + u_3 v_3,$$

and in particular the vector length satisfies

$$|\vec{v}|^2 = \langle \vec{v}, \vec{v} \rangle = v_1^2 + v_2^2 + v_3^2.$$

Analogy Between Vector and Fourier Expansions

Vectors:

$$\vec{v} = \sum_{n=1}^3 v_n \vec{e}_n,$$
$$v_n = \langle \vec{v}, \vec{e}_n \rangle$$

Fourier Series:

$$h(t) = \sum_{n=-\infty}^{\infty} H_n e^{-in\omega_0 t},$$
$$H_n = \frac{1}{T} \int_{t_0}^{t_0+T} h(t) e^{in\omega_0 t} dt$$

The analogy can be made precise by defining an inner product between T – periodic functions: $\langle g(t), h(t) \rangle = \frac{1}{T} \int_{t_0}^{t_0+T} g(t) \overline{h(t)} dt$

Then, the basis functions $\phi_n(t) = e^{-in\omega_0 t}$ play the role of the $\vec{e}_n$,

and we have $h(t) = \sum_{n=-\infty}^{\infty} H_n \phi_n(t), \quad H_n = \langle h(t), \phi_n(t) \rangle$

Parseval's Theorem

Vectors:

$$\vec{v} = \sum_{n=1}^3 v_n \vec{e}_n,$$

$$v_n = \langle \vec{v}, \vec{e}_n \rangle$$

$$|\vec{v}|^2 = v_1^2 + v_2^2 + v_3^2$$

Fourier Series:

$$h(t) = \sum_{n=-\infty}^{+\infty} H_n e^{-in\omega_0 t},$$

$$H_n = \frac{1}{T} \int_{t_0}^{t_0+T} h(t) e^{in\omega_0 t} dt = \langle h(t), e^{-in\omega_0 t} \rangle$$

$$\|h\|^2 = \frac{1}{T} \int_{t_0}^{t_0+T} |h(t)|^2 dt = \sum_{n=-\infty}^{\infty} |H_n|^2$$

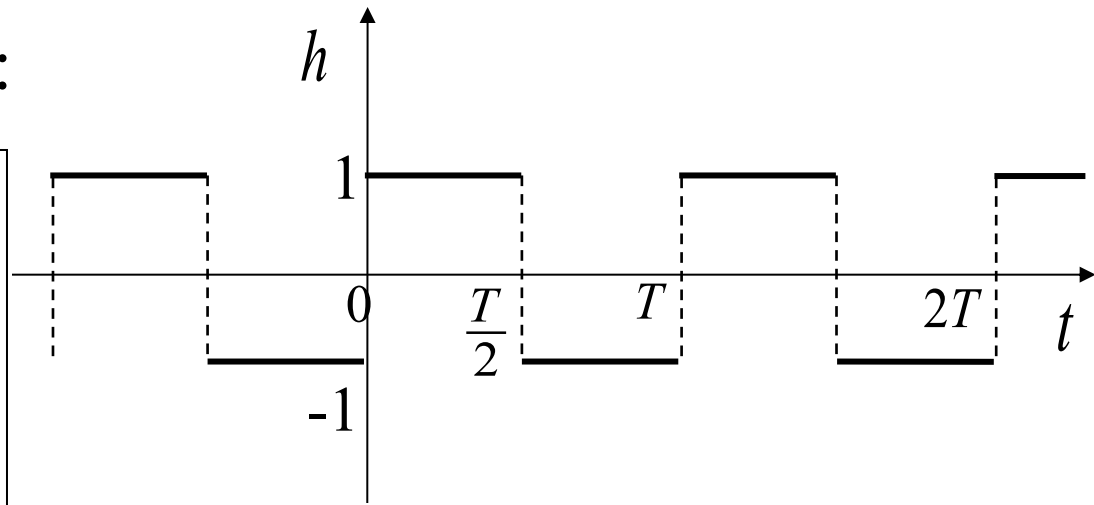
PARSEVAL'S RELATION

Proof:

$$\begin{aligned} \frac{1}{T} \int_{t_0}^{t_0+T} |h(t)|^2 dt &= \frac{1}{T} \int_{t_0}^{t_0+T} h(t) \overline{h(t)} dt = \frac{1}{T} \int_{t_0}^{t_0+T} h(t) \overline{\sum_{n=-\infty}^{\infty} H_n e^{-in\omega_0 t}} dt \\ &= \sum_{n=-\infty}^{\infty} \overline{H_n} \underbrace{\frac{1}{T} \int_{t_0}^{t_0+T} h(t) e^{in\omega_0 t} dt}_{H_n} = \sum_{n=-\infty}^{\infty} |H_n|^2 \end{aligned}$$

Example: Square Wave:

$$\begin{aligned} h(t) &= 2 \sum_{n=1}^{\infty} b_n \sin(n\omega_0 t) \\ &= \sum_{\substack{n=1 \\ n \text{ odd}}}^{\infty} \frac{4}{n\pi} \sin(n\omega_0 t) \end{aligned}$$



$$\frac{1}{T} \int_0^T h(t)^2 dt = 2 \sum_{\substack{n=1 \\ n \text{ odd}}}^{\infty} b_n^2 = 2 \sum_{\substack{n=1 \\ n \text{ odd}}}^{\infty} \left(\frac{2}{n\pi} \right)^2 = \frac{8}{\pi^2} \sum_{\substack{n=1 \\ n \text{ odd}}}^{\infty} \frac{1}{n^2}$$

Since $h(t)^2 \equiv 1$, the left-hand side is easily found to be 1.

Therefore, we conclude that $\frac{\pi^2}{8} = \sum_{\substack{n=1 \\ n \text{ odd}}}^{+\infty} \frac{1}{n^2} = 1 + \frac{1}{9} + \frac{1}{25} + \dots$

Try to check this with your calculator!

Fourier Transform

Fourier Series as the Period Increases

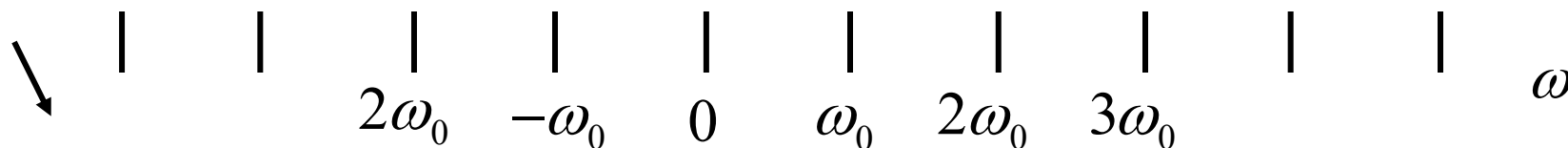
Consider the series $h(t) = \sum_{n=-\infty}^{\infty} H_n e^{-in\omega_0 t}$, for period $T = \frac{2\pi}{\omega_0}$

The frequencies involved in the expansion are $\{n\omega_0\}_{n=-\infty}^{+\infty}$.



Now consider a function with twice the period, $T = 2T$. Then

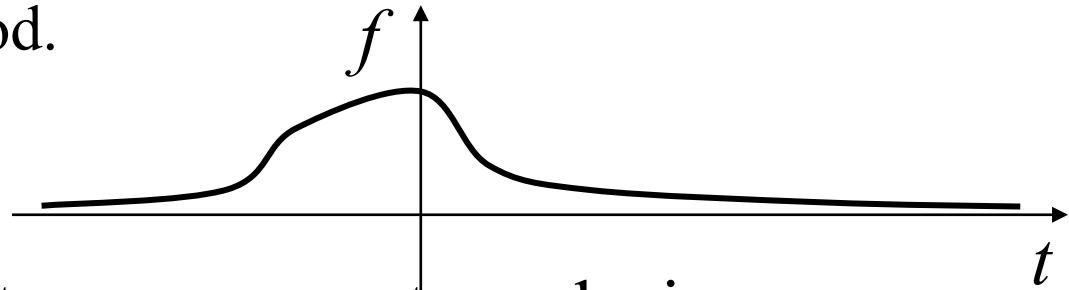
$\omega_0 = \frac{\omega_0}{2}$. The frequencies involved in the expansion are $\{n\omega_0\}_{n=-\infty}^{+\infty}$



As the period gets longer, we get a more dense set of frequencies. In the limit: include all frequencies.

Fourier Transform

We wish to extend the Fourier series concept to a non-periodic $h(t)$. Intuitively: $h(t)$ takes infinitely long to repeat itself, so we think of it as a function of infinite period.



The argument in the last page suggests replacing

$$h(t) = \sum_{n=-\infty}^{\infty} H_n e^{-in\omega_0 t} \text{ by an integral where } T \rightarrow \infty$$

Definition: The Fourier transform of time-domain function $h(t)$, $t \in (-\infty, +\infty)$ is the function of frequency $f \in (-\infty, +\infty)$ defined by the integral $H(f) = \mathcal{F}[h(t)] = \int_{-\infty}^{\infty} h(t) e^{2\pi i f t} dt$, assuming it converges. (Here we define $f = \omega/2\pi$)

Inverse Fourier Transform

In Fourier series, we can reconstruct the function from the Fourier coefficients via $h(t) = \sum_{n=-\infty}^{\infty} H_n e^{-in\omega_0 t}$.

Similarly, for Fourier transforms we can reconstruct the function using the inverse formula

$$h(t) = \int_{-\infty}^{\infty} H(f) e^{-2\pi i f t} df$$

A complete derivation is mathematically involved, but we can sketch a proof based on Fourier series case, letting the period go to infinity.

Fourier Integral Theorem

Fourier transform and its inverse are based on the Fourier Integral Theorem:

If $h(t)$ is integrable on $(-\infty, \infty)$ and satisfies the Dirichlet conditions in arbitrary intervals in $(-\infty, \infty)$, then

$$\int_{-\infty}^{\infty} \left[\int_{-\infty}^{\infty} h(\tau) e^{2\pi i f(\tau - t)} d\tau \right] df = \frac{1}{2} [h(t+) + h(t-)]$$

Summary

Fourier Series
(T-periodic functions)

$$H_n = \frac{1}{T} \int_{t_0}^{t_0+T} h(t) e^{2\pi i n f_0 t} dt$$

$$h(t) = \sum_{n=-\infty}^{\infty} H_n e^{-2\pi i n f_0 t}$$

Fourier Transforms

$$H(f) = \int_{-\infty}^{\infty} h(t) e^{2\pi i f t} dt$$

$$h(t) = \int_{-\infty}^{\infty} H(f) e^{-2\pi i f t} df$$

Properties of the Fourier Transform

$$H(f) = \mathcal{F}[h(t)] = \int_{-\infty}^{\infty} h(t) e^{2\pi i f t} dt$$

$$h(t) = \mathcal{F}^{-1}[H(f)] = \int_{-\infty}^{\infty} H(f) e^{-2\pi i f t} df$$

1) **Linearity.** $\mathcal{F}[\alpha h(t) + \beta g(t)] = \alpha \mathcal{F}[h(t)] + \beta \mathcal{F}[g(t)]$

Follows directly from linearity of the integral.

2) $\mathcal{F}[h(-t)] = H(-f)$.

Proof :
$$\int_{-\infty}^{\infty} h(-t) e^{2\pi i f t} dt = \int_{-\infty}^{\infty} h(\sigma) e^{-2\pi i f \sigma} d\sigma = H(-f)$$

$$\sigma = -t$$

3) If $h(t)$ is real-valued, then $H(-f) = \overline{H(f)}$

$$\text{Proof : } \int_{-\infty}^{\infty} h(t) e^{-2\pi i f t} dt = \int_{-\infty}^{\infty} \overline{h(t) e^{2\pi i f t}} dt = \overline{\int_{-\infty}^{\infty} h(t) e^{2\pi i f t} dt}$$

4a) $h(t)$ real and even ($h(t) = h(-t)$) $\Rightarrow H(f)$ real and even.

$$\text{Proof: } H(f) = \mathcal{F}[h(t)] = \mathcal{F}[h(-t)] = H(-f) = \overline{H(f)}$$

2) 3)

4b) $h(t)$ real and odd ($h(t) = -h(-t)$) $\Rightarrow H(f)$ purely imaginary and odd.

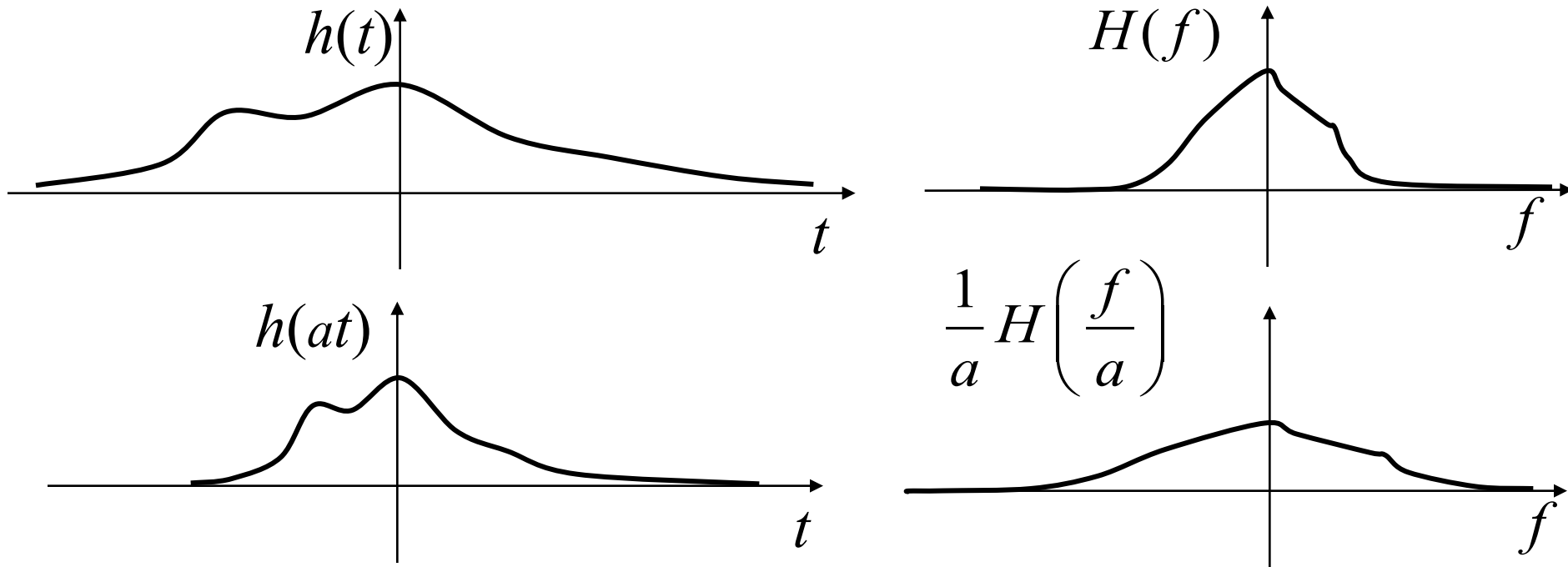
$$\text{Proof: } H(f) = -\mathcal{F}[h(-t)] = -H(-f) = -\overline{H(f)}$$

1) 2) 3)

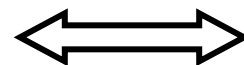
5) Time scaling: $\mathcal{F}[h(at)] = \frac{1}{|a|} H\left(\frac{f}{a}\right)$

Proof ($a > 0$): (Work out the proof for $a < 0$ yourself)

$$\int_{-\infty}^{\infty} h(at) e^{2\pi i f t} dt = \int_{-\infty}^{\infty} h(\sigma) e^{2\pi i \frac{f}{a} \sigma} \frac{d\sigma}{a} = \frac{1}{a} H\left(\frac{f}{a}\right).$$



e.g, $a > 1$ Compress time

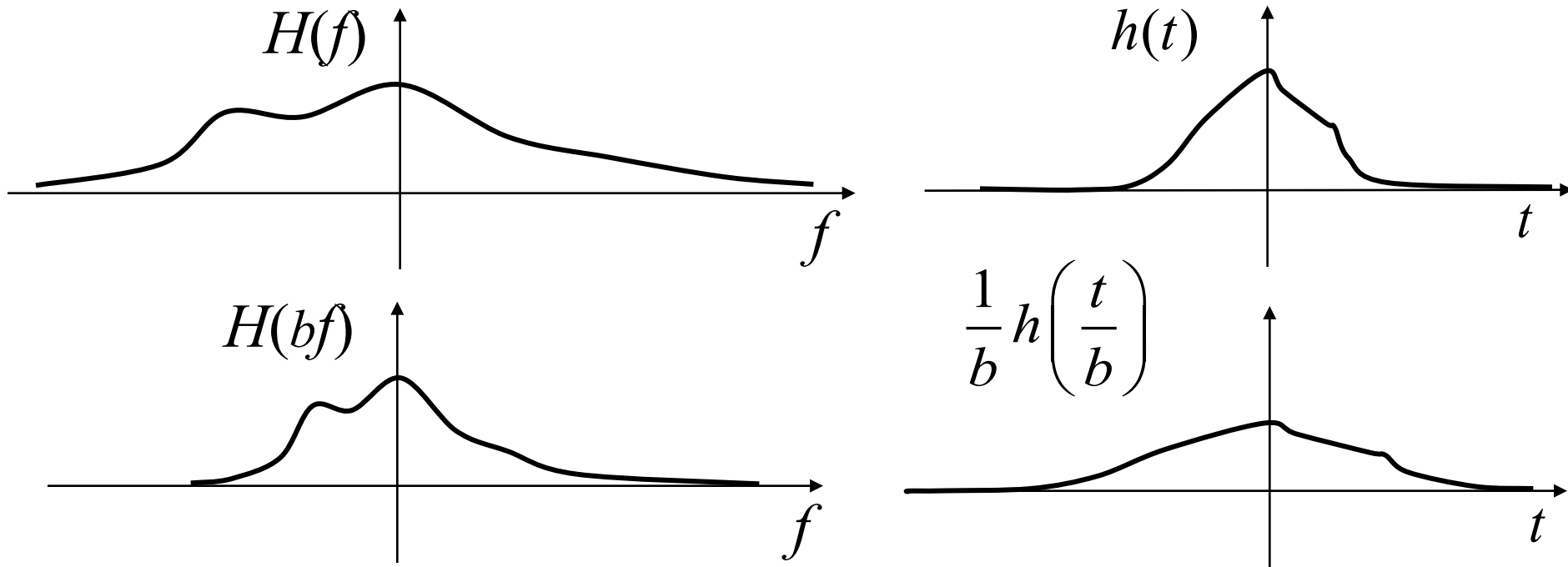


Stretch frequency,
and scale down.

6) Frequency scaling: $\mathcal{F}^{-1}[H(bf)] = \frac{1}{|b|} h\left(\frac{t}{b}\right)$

Proof ($b > 0$): (Work out the proof for $b < 0$ yourself)

$$\int_{-\infty}^{\infty} H(bf) e^{-2\pi i f t} df = \int_{-\infty}^{\infty} H(\sigma) e^{-2\pi i \sigma \frac{t}{b}} \frac{d\sigma}{b} = \frac{1}{b} h\left(\frac{t}{b}\right).$$



e.g., $b > 1$ Compress frequency $\longleftrightarrow$ Stretch time,
and scale down.

7) Time shifting: $\mathcal{F}[h(t-t_0)] = e^{2\pi i f t_0} H(f)$

$$\text{Proof : } \int_{-\infty}^{\infty} h(t-t_0) e^{2\pi i f t} dt = \int_{-\infty}^{\infty} h(\sigma) e^{2\pi i f (\sigma+t_0)} d\sigma$$

$\sigma = t - t_0$

Remark: no change in magnitude $\left| e^{2\pi i f t_0} H(f) \right| = |H(f)|$,
but we add a phase that is linear in f

8) Frequency shifting: $\mathcal{F}\left[e^{-2\pi i f_0 t} h(t)\right] = H(f - f_0)$

$$\text{Proof : } \int_{-\infty}^{\infty} e^{-2\pi i f_0 t} h(t) e^{2\pi i f t} dt = \int_{-\infty}^{\infty} h(t) e^{2\pi i (f - f_0) t} dt$$

Note: dual of the delay property

Parseval Theorem

$$\int_{-\infty}^{\infty} |h(t)|^2 dt = \int_{-\infty}^{\infty} |H(f)|^2 df$$

Integral square is preserved between time and frequency.

$$\begin{aligned} \text{Proof : } \int_{-\infty}^{\infty} |h(t)|^2 dt &= \int_{-\infty}^{\infty} h(t) \overline{h(t)} dt \\ &= \int_{-\infty}^{\infty} h(t) \overline{\left(\int_{-\infty}^{\infty} H(f) e^{-2\pi i f t} df \right)} dt \\ &= \int_{-\infty}^{\infty} \overline{H(f)} \underbrace{\left(\int_{-\infty}^{\infty} h(t) e^{2\pi i f t} dt \right)}_{H(f)} df = \int_{-\infty}^{\infty} |H(f)|^2 df \end{aligned}$$

Convolution

Def:

$$y(t) = h(t) * g(t) = \int_{-\infty}^{\infty} h(t - \tau) g(\tau) d\tau$$

Theorem: Let $G(f) = \mathcal{F}[g(t)]$, $H(f) = \mathcal{F}[h(t)]$, $Y(f) = \mathcal{F}[y(t)]$
(assume they exist). Then $\boxed{Y(f) = H(f)G(f)}$ for all f

The Fourier transform takes convolutions into products.

Proof: $Y(f) = \int_{-\infty}^{\infty} y(t) e^{2\pi i f t} dt = \int_{-\infty}^{\infty} \left[\int_{-\infty}^{\infty} h(t - \tau) g(\tau) d\tau \right] e^{2\pi i f t} dt$

Change order
of integration.

$$= \int_{-\infty}^{\infty} \left[\int_{-\infty}^{\infty} h(t - \tau) e^{2\pi i f t} dt \right] g(\tau) d\tau$$

$$\stackrel{\sigma = t - \tau}{=} \int_{-\infty}^{\infty} \left[\int_{-\infty}^{\infty} h(\sigma) e^{2\pi i f (\sigma + \tau)} d\sigma \right] g(\tau) d\tau$$

$$= \int_{-\infty}^{\infty} h(\sigma) e^{2\pi i f \sigma} d\sigma \int_{-\infty}^{\infty} e^{2\pi i f \tau} g(\tau) d\tau = H(f) G(f)$$

Examples of Convolution

Throwing two dice, win if the sum of the outcomes is c

The dice are in general loaded dice

Let $p_1(x)$ ($p_2(x)$) be the probability of getting x throwing dice 1 (2).

(The dice are fair dice if $p_i(x) = 1/6$)

Then probability of winning = $\sum_a p_1(a)p_2(c-a)$

In experiments, if want to measure something described by $g(t)$,
but subjected to experimental resolution $h(t)$, then the expt. result is

$$(g * h)(t)$$

Use convolution to get $g(t)$.

Discrete Time Fourier Transform

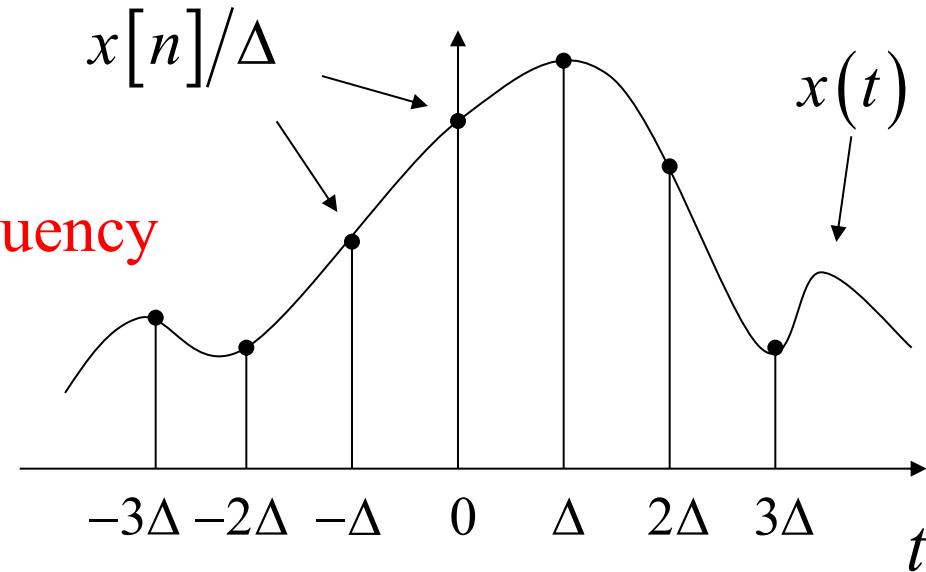
DTFT

Sampling

$x(t)$ is sampled $\rightarrow x[n] = x(n\Delta)\Delta \quad n = \dots -3, -2, -1, 0, 1, 2, 3, \dots$

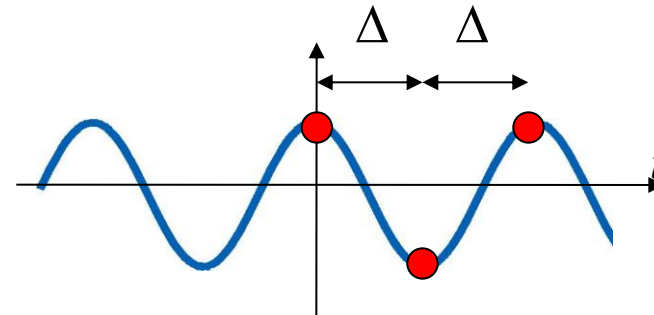
$\frac{1}{\Delta}$: Sampling rate

$f_c \equiv \frac{1}{2\Delta}$: **Nyquist critical frequency**



Signal with frequency f_c

$\rightarrow$ Two sample points per cycle



FT of Sample Sequence

Consider the sequence of sampled data $\sum_{n=-\infty}^{\infty} x[n] \delta(t - n\Delta)$

By Fourier transform

$$X(f) = \int_{-\infty}^{\infty} \left(\sum_{n=-\infty}^{\infty} x[n] \delta(t - n\Delta) \right) e^{2\pi i f t} dt = \sum_{n=-\infty}^{\infty} x[n] e^{2\pi i f n \Delta}$$

Note: Infinite discrete $t = n\Delta$

→ f is continuous and $X(f)$ is periodic with period $1/\Delta = 2f_c$

Recall: Fourier series, infinite discrete $f = nf_0$

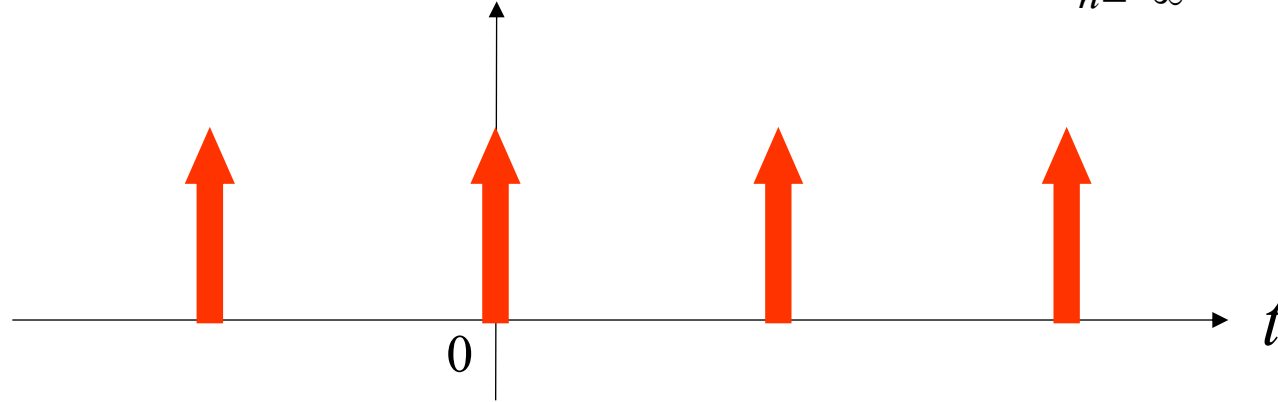
→ t is continuous and $x(t)$ is periodic with period $1/f_0$

Duality of f and t

Question: Can one determine $x(t)$ from the sampled $x[n]$?

Dirac Comb

Consider the function of Δ times the Dirac comb: $h(t) = \Delta \sum_{n=-\infty}^{\infty} \delta(t - n\Delta)$



This is a periodic function with period Δ

→ Expand by Fourier series, the Fourier coefficients are

$$\frac{1}{\Delta} \int_{-\Delta/2}^{\Delta/2} h(t) e^{2\pi i n t / \Delta} dt = \frac{1}{\Delta} \int_{-\Delta/2}^{\Delta/2} \Delta \delta(t) e^{2\pi i n t / \Delta} dt = 1$$

$$\Rightarrow h(t) = \sum_{n=-\infty}^{\infty} e^{-2\pi i n t / \Delta}$$

Relation of FT of Signal and Sample

The sample points are given by $\sum_{n=-\infty}^{\infty} x[n] \delta(t - n\Delta) = x(t)h(t)$

$$\begin{aligned} X(f) &= \int_{-\infty}^{\infty} x(t)h(t)e^{2\pi ift} dt \\ &= \int_{-\infty}^{\infty} \left(\sum_{n=-\infty}^{\infty} e^{-2\pi int/\Delta} \right) x(t)e^{2\pi ift} dt \\ &= \sum_{n=-\infty}^{\infty} \int_{-\infty}^{\infty} x(t)e^{2\pi i(f-2nf_c)t} dt \\ &= \sum_{n=-\infty}^{\infty} \tilde{X}(f - 2nf_c) \end{aligned}$$

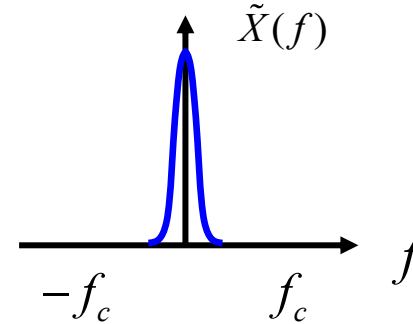
Band-Width Limited Signals

If $\tilde{X}(f) = \mathcal{F}\{x(t)\}$ is zero for frequencies exceeding some cutoff frequency, we say that the signal $x(t)$ is bandwidth limited

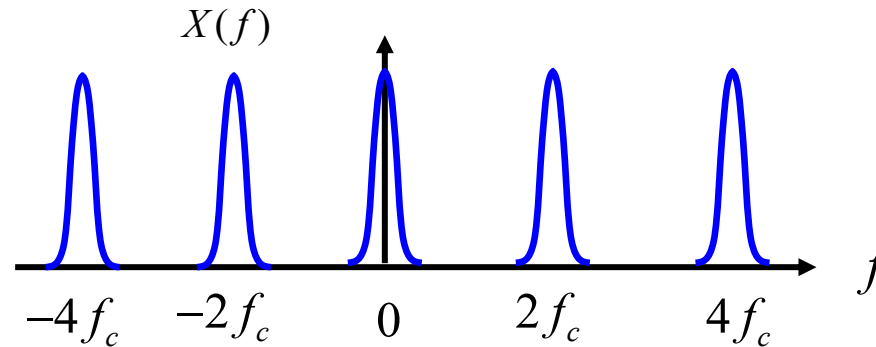
For example, a signal $x(t)$, sampled at an interval Δ , may satisfy $\tilde{X}(f) = 0 \quad \forall |f| \geq f_c$

Band-Width Limited Signals

If $\tilde{X}(f)$ is bandwidth limited by f_c



Then there is no overlap
of $\tilde{X}(f)$ in $X(f)$



$$X(f) = \sum_{n=-\infty}^{\infty} \tilde{X}(f - 2nf_c) \Rightarrow \tilde{X}(f) = \begin{cases} X(f) & |f| < f_c \\ 0 & |f| > f_c \end{cases}$$

$\tilde{X}(f)$ completely determined by $X(f) \Rightarrow x(t)$ completely determined by $x[n]$

Nyquist–Shannon Sampling Theorem

If a signal $x(t)$, sampled at an interval Δ ,
is bandwidth limited to frequency smaller
in magnitude than f_c , i.e., $\tilde{X}(f) = 0 \quad \forall |f| \geq f_c$,
then $x(t)$ is completely determined by its samples $x[n]$:

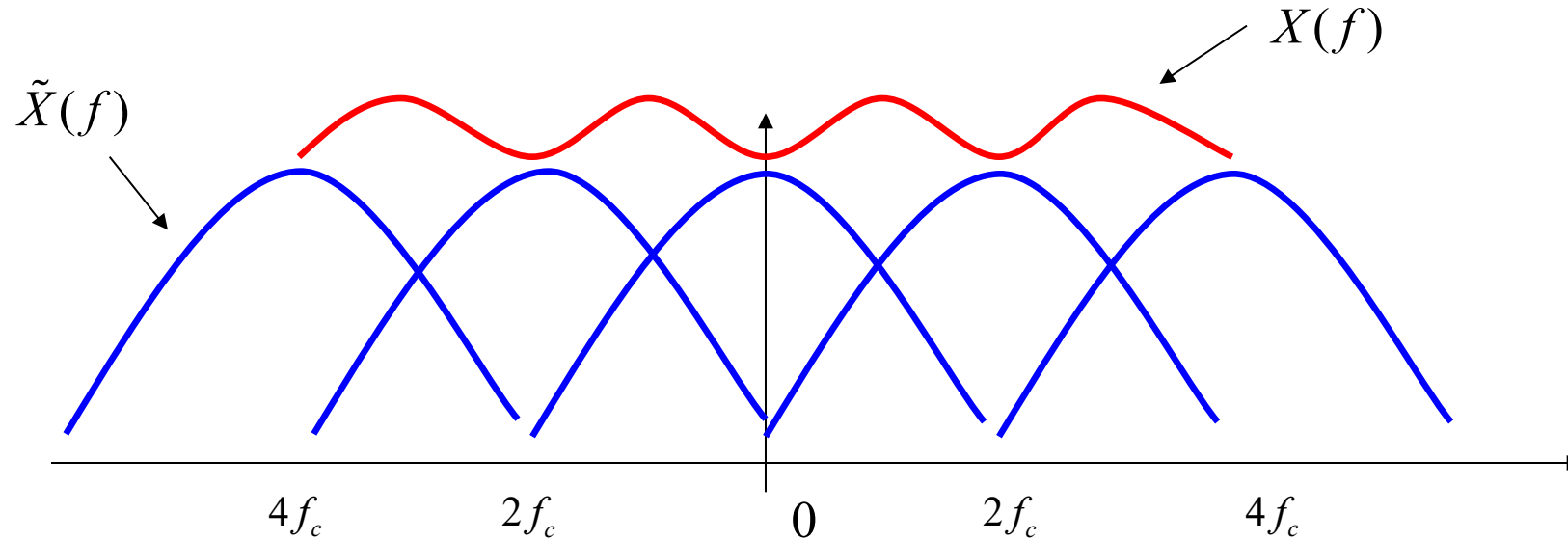
$$x(t) = \sum_{n=-\infty}^{\infty} x[n] \frac{\sin[2\pi f_c(t - n\Delta)]}{\pi(t - n\Delta)}$$

Proof

$$\begin{aligned} X(f) &= \int_{-\infty}^{\infty} \left[\sum_{n=-\infty}^{\infty} x[n] \delta(t - n\Delta) \right] e^{2\pi i f t} dt \\ &= \sum_{n=-\infty}^{\infty} x[n] \int_{-\infty}^{\infty} \delta(t - n\Delta) e^{2\pi i f t} dt = \sum_{n=-\infty}^{\infty} x[n] e^{2\pi i f n \Delta} \\ \therefore x(t) &= \int_{-f_c}^{f_c} \left(\sum_{n=-\infty}^{\infty} x[n] e^{2\pi i f n \Delta} \right) e^{-2\pi i f t} df \\ &= \sum_{n=-\infty}^{\infty} x[n] \int_{-f_c}^{f_c} e^{2\pi i f (n\Delta - t)} df = \sum_{n=-\infty}^{\infty} \frac{x[n]}{2\pi i (n\Delta - t)} \left[e^{2\pi i f (n\Delta - t)} \right]_{-f_c}^{f_c} \\ &= \sum_{n=-\infty}^{\infty} x[n] \frac{e^{2\pi i f_c (n\Delta - t)} - e^{-2\pi i f_c (n\Delta - t)}}{2\pi i (n\Delta - t)} \\ &= \sum_{n=-\infty}^{\infty} x[n] \frac{\sin 2\pi f_c (t - n\Delta)}{\pi (t - n\Delta)} \end{aligned}$$

Aliasing

What if $\tilde{X}(f)$ is not bandwidth limited?

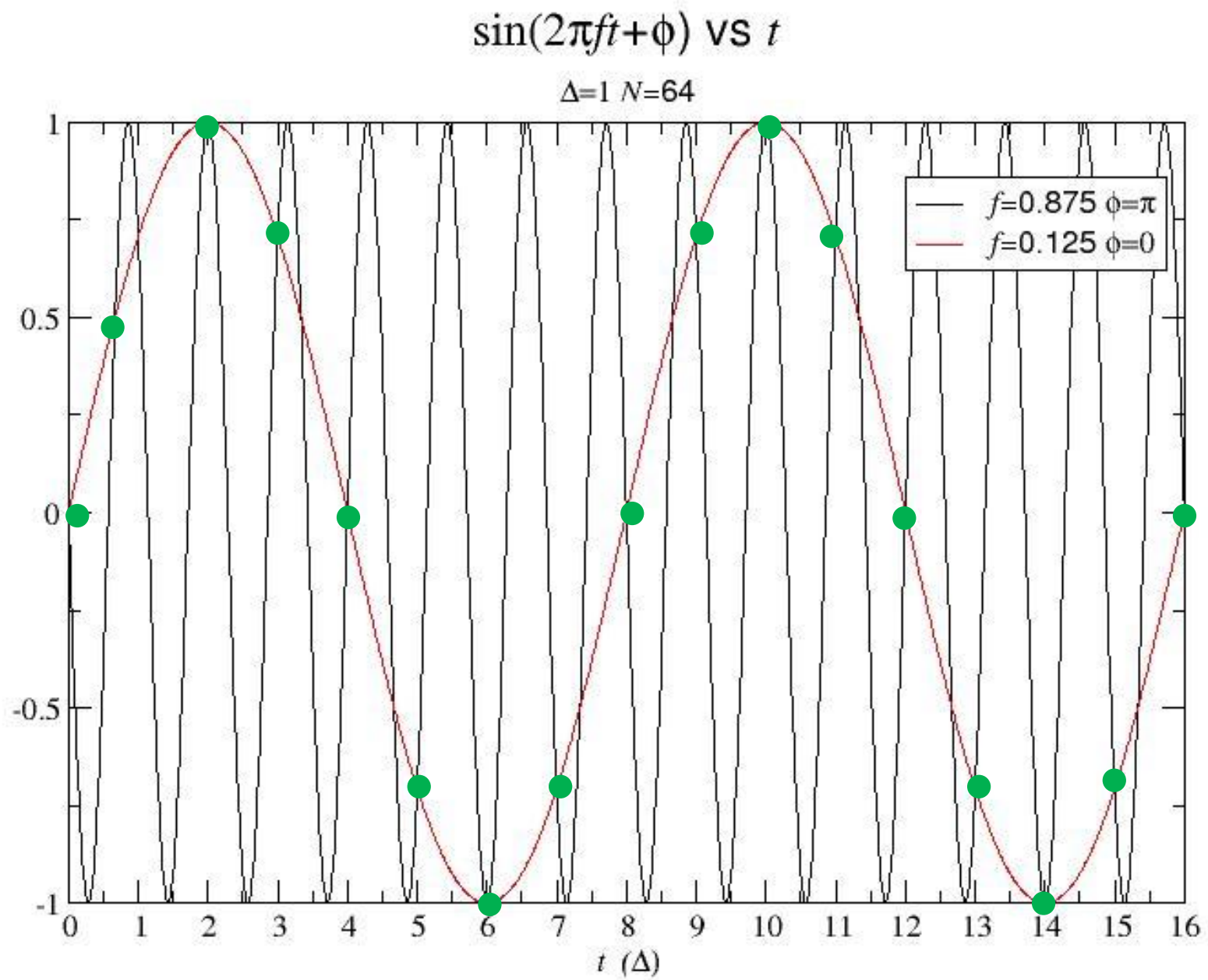


Frequencies outside $-f_c < f < f_c$ are spuriously moved into the region

$$X(f) = \sum_{n=-\infty}^{\infty} \tilde{X}(f - 2nf_c)$$

Physical meaning: frequencies differ by a multiple of $2f_c$
cannot be distinguished by the sampling

$$X(f_0) = \left[\tilde{X}(f_0) + \tilde{X}(f_0 \pm 2f_c) + \tilde{X}(f_0 \pm 4f_c) + \cdots \right]$$



How to Ensure?

- Know in advance (e.g., from physics) that the signal is bandwidth limited
- Signal passed through a frequency filter
- Check if $\tilde{X}(f) \rightarrow 0$ at the boundaries

Discrete Fourier Transform (DFT)

- In real-life problems, usually we only have finite number of discrete data
- In this case, not only are the FTs periodic in f , but the frequencies are also discrete
- Discrete + Periodic $\rightarrow$ Finite number of distinct frequencies
- For N discrete time-domain data, we only need N discrete frequencies
- The corresponding inverse Fourier transform can correctly give back the sampled data
- If the inverse FT is extended to represent data as well outside the range, they may or may not be correct (It depends on the system under study, e.g., there may be no other data apart from the set you have at all)
- In particular if your data is really the set of an infinite periodic sequence sampled in one period, then the DFT correctly represent the whole periodic sequence

Discrete Fourier Transform (DFT)

N data $h[n]$ $n = 0, 1, 2, \dots, N-1$

DFT:

$$H[k] = \sum_{n=0}^{N-1} h[n] e^{2\pi i k n / N}$$

$$h[n] = \frac{1}{N} \sum_{k=0}^{N-1} H[k] e^{-2\pi i k n / N}$$

Proof

$$\begin{aligned}\text{Consider } \frac{1}{N} \sum_{k=0}^{N-1} e^{-2\pi i k n / N} e^{2\pi i k m / N} &= \frac{1}{N} \sum_{k=0}^{N-1} e^{2\pi i k (m-n) / N} \\ &= \frac{1}{N} \left[1 + \alpha + \alpha^2 + \cdots + \alpha^{N-1} \right]\end{aligned}$$

$$\text{where } \alpha = e^{2\pi i (m-n) / N}$$

$$\text{If } m = n \quad \frac{1}{N} \sum_{k=0}^{N-1} e^{2\pi i k (m-n) / N} = 1$$

$$\text{If } m \neq n$$

$$\begin{aligned}\frac{1}{N} \sum_{k=0}^{N-1} e^{2\pi i k (m-n) / N} &= \frac{1}{N} \frac{1 - \alpha^N}{1 - \alpha} = \frac{1}{N} \frac{1 - e^{2\pi i (m-n)}}{1 - e^{2\pi i (m-n) / N}} \\ &= 0\end{aligned}$$

$$\therefore \frac{1}{N} \sum_{k=0}^{N-1} e^{-2\pi i k n / N} e^{2\pi i k m / N} = \delta_{mn}$$

Proof

$$\begin{aligned}h[n] &= \sum_{m=0}^{N-1} h[m] \delta_{mn} \\&= \sum_{m=0}^{N-1} h[m] \left[\frac{1}{N} \sum_{k=0}^{N-1} e^{-2\pi i k n / N} e^{2\pi i k m / N} \right] \\&= \frac{1}{N} \sum_{k=0}^{N-1} \left[\sum_{m=0}^{N-1} h[m] e^{2\pi i k m / N} \right] e^{-2\pi i k n / N} \\ \therefore h[n] &= \frac{1}{N} \sum_{k=0}^{N-1} H[k] e^{-2\pi i k n / N} \quad \leftrightarrow \quad H[k] = \sum_{m=0}^{N-1} h[m] e^{2\pi i k m / N}\end{aligned}$$

Parseval's Theorem

$$\sum_{k=0}^{N-1} |h[k]|^2 = \frac{1}{N} \sum_{n=0}^{N-1} |H[n]|^2$$

Proof:

$$\begin{aligned} \frac{1}{N} \sum_{n=0}^{N-1} |H[n]|^2 &= \frac{1}{N} \sum_{n=0}^{N-1} H[n] \overline{H[n]} \\ &= \frac{1}{N} \sum_{n=0}^{N-1} \left(\sum_{p=0}^{N-1} h[p] e^{2\pi i n p / N} \right) \overline{\left(\sum_{q=0}^{N-1} h[q] e^{2\pi i n q / N} \right)} \\ &= \frac{1}{N} \sum_{n=0}^{N-1} \left(\sum_{p=0}^{N-1} h[p] e^{2\pi i n p / N} \right) \left(\sum_{q=0}^{N-1} \overline{h[q]} e^{-2\pi i n q / N} \right) \\ &= \sum_{p=0}^{N-1} \sum_{q=0}^{N-1} h[p] \overline{h[q]} \left(\frac{1}{N} \sum_{n=0}^{N-1} e^{2\pi i n (p-q) / N} \right) \\ &= \sum_{p=0}^{N-1} \sum_{q=0}^{N-1} h[p] \overline{h[q]} \delta_{pq} \\ &= \sum_{p=0}^{N-1} |h[p]|^2 \end{aligned}$$

Period of the DFT

$$H[k] = \sum_{n=0}^{N-1} h[n] e^{2\pi i k n / N} \text{ so defined, have period } N: \boxed{H[k+N] = H[k]}$$

$$\text{Proof: } H[k+N] = \sum_{n=0}^{N-1} h[n] e^{2\pi i n (k+N) / N} = \sum_{n=0}^{N-1} h[n] e^{2\pi i k n / N} e^{2\pi i n} = \sum_{n=0}^{N-1} h[n] e^{2\pi i k n / N} = H[k]$$

$$\text{Inverse DFT also gives a periodic sampling data with period } N: \boxed{h[n+N] = h[n]}$$

i.e., if $h[n] = \frac{1}{N} \sum_{k=0}^{N-1} H[k] e^{-2\pi i k n / N}$ is extended to represent data outside the range,

one has a sequence of data with period N

$$\text{Proof: } h[n+N] = \sum_{k=0}^{N-1} H[k] e^{-2\pi i k (n+N) / N} = \sum_{k=0}^{N-1} H[k] e^{-2\pi i k n / N} e^{-2\pi i k} = \sum_{k=0}^{N-1} H[k] e^{-2\pi i k n / N} = h[n]$$

Choice of Frequency Range

Any N frequencies k with $k(\bmod N) = 0, 1, 2, 3, \dots, N-1$ give *complete* and *equivalent* description of the N discrete data $h[n]$. $H[k]$ and $H[k + pN]$ are indistinguishable by the sampling

We will therefore assume $k = 0, 1, 2, \dots, N-1$

In practice, one may know, say, from physics, that the signal is bandwidth limited with positive and negative components satisfying

$$-f_c = -\frac{1}{2\Delta} \leq f \leq \frac{1}{2\Delta} = f_c$$

In this case, one should remember that

- zero frequency $\rightarrow k = 0$
- positive frequencies $\rightarrow 1 \leq k \leq \lfloor N/2 \rfloor$
- negative frequencies $\rightarrow \lfloor N/2 \rfloor + 1 \leq k \leq N-1$

Two-Dimensional Fourier Transform

- Consider a two-variable function $h(x, y)$
- Suppose we have an $M \times N$ grid of data h_{mn}
- To carry out the 2D FT, we first perform an ordinary FT on each of the M rows by

$$H'_{ml} = \sum_{n=0}^{N-1} h_{mn} \exp\left(i \frac{2\pi ln}{N}\right)$$

- For each row m we now have N coefficients, one for each value of l
- Next we take the l -th coefficient in each of the M rows and Fourier transform these M values again to get

$$H_{kl} = \sum_{m=0}^{M-1} H'_{ml} \exp\left(i \frac{2\pi km}{M}\right)$$

Two-Dimensional Fourier Transform

- The overall transformation is

$$H_{kl} = \sum_{m=0}^{M-1} \sum_{n=0}^{N-1} h_{mn} \exp \left[2\pi i \left(\frac{km}{M} + \frac{ln}{N} \right) \right]$$

- The corresponding inverse transform is

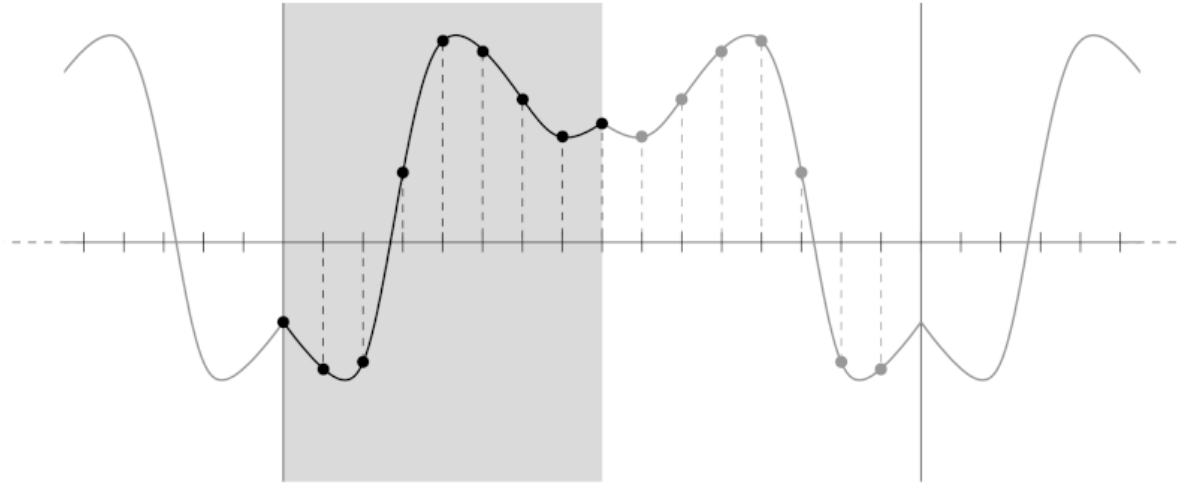
$$h_{mn} = \frac{1}{MN} \sum_{k=0}^{M-1} \sum_{l=0}^{N-1} H_{kl} \exp \left[-2\pi i \left(\frac{km}{M} + \frac{ln}{N} \right) \right]$$

Discrete Cosine Transform (DCT)

- DCT forms the mathematical basis for the computer image file format JPEG, a much more economic format instead of storing the data of every pixel
- JPEG format works by dividing the pixels in an image into blocks, performing DCTs on the blocks, then looking for Fourier coefficients that are small and can be discarded
- The remaining coefficients are stored in a file and when you view a picture, your computer reconstitutes the picture using the inverse transform
- MPEG is a variant of the same technique to compress moving pictures
- Similar schemes are used to compress music like MP3
- Because human ears are much less sensitive to high-frequency sounds when there are simultaneous loud low-frequency sounds, some high-frequency sounds are thrown away in MP3 in this case

Type I DCT

- Given N sampled data
- Construct the even-extension with $N + N - 2 = 2(N - 1)$ data



- $h[2N - n - 2] = h[n]$

Type I DCT

$$\begin{aligned} H[k] &= \sum_{n=0}^{2N-3} h[n] \exp\left(i \frac{2\pi kn}{2(N-1)}\right) \\ &= \sum_{n=0}^{N-1} h[n] \exp\left(i \frac{2\pi kn}{2(N-1)}\right) + \sum_{n=N}^{2N-3} h[n] \exp\left(i \frac{2\pi kn}{2(N-1)}\right) \\ &= \sum_{n=0}^{N-1} h[n] \exp\left(i \frac{2\pi kn}{2(N-1)}\right) + \sum_{n=N}^{2N-3} h[2N-n-2] \exp\left(-i \frac{2\pi k(2N-n-2)}{2(N-1)}\right) \end{aligned}$$

Change of notation of variable $2N-n-2 \rightarrow n$ in the second term:

$$\begin{aligned} H[k] &= \sum_{n=0}^{N-1} h[n] \exp\left(i \frac{2\pi kn}{2(N-1)}\right) + \sum_{n=1}^{N-2} h[n] \exp\left(-i \frac{2\pi kn}{2(N-1)}\right) \\ &= h[0] + h[N-1] \exp\left(i \frac{2\pi k(N-1)}{2(N-1)}\right) + \sum_{n=1}^{N-2} h[n] \left[\exp\left(i \frac{2\pi kn}{2(N-1)}\right) + \exp\left(-i \frac{2\pi kn}{2(N-1)}\right) \right] \\ &= h[0] + h[N-1] \exp(ik\pi) + 2 \sum_{n=1}^{N-2} h[n] \cos\left(\frac{\pi kn}{N-1}\right) \\ &= h[0] + (-1)^k h[N-1] + 2 \sum_{n=1}^{N-2} h[n] \cos\left(\frac{\pi kn}{N-1}\right) \end{aligned}$$

Type I DCT

It is easy to verify that (as expected):

$$\begin{aligned} H[2N - k - 2] &= h[0] + (-1)^{2N-k-2} h[N-1] + 2 \sum_{n=1}^{N-2} h[n] \cos\left(\frac{\pi(2N-k-2)n}{N-1}\right) \\ &= h[0] + (-1)^k h[N-1] + 2 \sum_{n=1}^{N-2} h[n] \cos\left(\frac{\pi kn}{N-1}\right) \\ &= H[k] \end{aligned}$$

Inverse Type I DCT

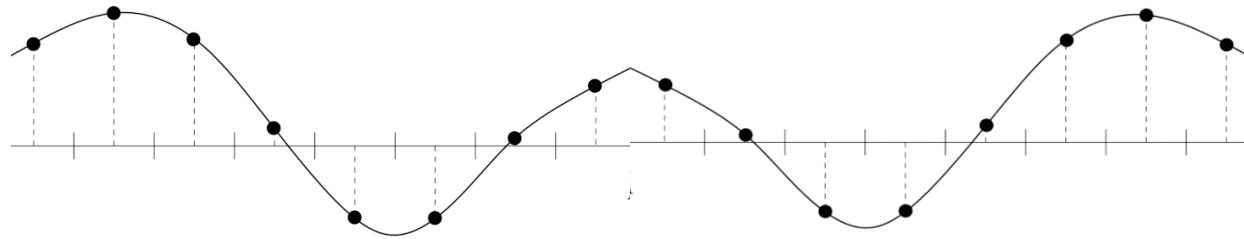
$$\begin{aligned}h[n] &= \frac{1}{2(N-1)} \sum_{k=0}^{2N-3} H[k] \exp\left(-i \frac{2\pi kn}{2(N-1)}\right) \\&= \frac{1}{2(N-1)} \sum_{k=0}^{N-1} H[k] \exp\left(-i \frac{2\pi kn}{2(N-1)}\right) + \frac{1}{2(N-1)} \sum_{k=N}^{2N-3} H[k] \exp\left(-i \frac{2\pi kn}{2(N-1)}\right) \\&= \frac{1}{2(N-1)} \sum_{k=0}^{N-1} H[k] \exp\left(-i \frac{2\pi kn}{2(N-1)}\right) + \frac{1}{2(N-1)} \sum_{k=N}^{2N-3} H[2N-k-2] \exp\left(i \frac{2\pi n(2N-k-2)}{2(N-1)}\right)\end{aligned}$$

Change of notation of variable $2N-k-2 \rightarrow k$ in the second term:

$$\begin{aligned}h[n] &= \frac{1}{2(N-1)} \sum_{k=0}^{N-1} H[k] \exp\left(-i \frac{2\pi kn}{2(N-1)}\right) + \frac{1}{2(N-1)} \sum_{k=1}^{N-2} H[k] \exp\left(i \frac{2\pi kn}{2(N-1)}\right) \\&= \frac{1}{2(N-1)} \left[H[0] + H[N-1] \exp\left(-i \frac{2\pi n(N-1)}{2(N-1)}\right) + \sum_{k=1}^{N-2} H[k] \left[\exp\left(i \frac{2\pi kn}{2(N-1)}\right) + \exp\left(-i \frac{2\pi kn}{2(N-1)}\right) \right] \right] \\&= \frac{1}{2(N-1)} \left[H[0] + H[N-1] \exp(-in\pi) + 2 \sum_{k=1}^{N-2} H[k] \cos\left(\frac{\pi kn}{N-1}\right) \right] \\&= \frac{1}{2(N-1)} \left[H[0] + (-1)^n H[N-1] + 2 \sum_{k=1}^{N-2} H[k] \cos\left(\frac{\pi kn}{N-1}\right) \right]\end{aligned}$$

Type II DCT

- Sample points are in the middle of the sample intervals rather than at their ends
- Construct the even-extension with $2N$ data



- $$h[2N - n - 1] = h[n]$$

Type II DCT

$$\begin{aligned} H[k] &= \sum_{n=0}^{2N-1} h[n] \exp\left(i \frac{2\pi kn}{2N}\right) \\ &= \sum_{n=0}^{N-1} h[n] \exp\left(i \frac{2\pi kn}{2N}\right) + \sum_{n=N}^{2N-1} h[n] \exp\left(i \frac{2\pi kn}{2N}\right) \\ &= \sum_{n=0}^{N-1} h[n] \exp\left(i \frac{2\pi kn}{2N}\right) + \sum_{n=N}^{2N-1} h[2N-n-1] \exp\left(-i \frac{2\pi k(2N-n)}{2N}\right) \end{aligned}$$

Change of notation of variable $2N-n-1 \rightarrow n$ in the second term:

$$\begin{aligned} H[k] &= \sum_{n=0}^{N-1} h[n] \exp\left(i \frac{2\pi kn}{2N}\right) + \sum_{n=0}^{N-1} h[n] \exp\left(-i \frac{2\pi k(n+1)}{2N}\right) \\ &= \exp\left(-i \frac{\pi k}{2N}\right) \left[\sum_{n=0}^{N-1} h[n] \exp\left(i \frac{2\pi k(n+1/2)}{2N}\right) + \sum_{n=0}^{N-1} h[n] \exp\left(-i \frac{2\pi k(n+1/2)}{2N}\right) \right] \\ &= 2 \exp\left(-i \frac{\pi k}{2N}\right) \sum_{n=0}^{N-1} h[n] \cos\left(\frac{\pi k(n+1/2)}{N}\right) \end{aligned}$$

Type II DCT

The prefactor is conventionally absorbed into the Fourier coefficients by defining

$$\tilde{H}[k] = \frac{1}{2} H[k] \exp\left(i \frac{\pi k}{2N}\right) = \sum_{n=0}^{N-1} h[n] \cos\left(\frac{\pi k (n+1/2)}{N}\right)$$

which are real if $h[n]$ are real.

Also

$$\begin{aligned} \tilde{H}[2N-k] &= \sum_{n=0}^{N-1} h[n] \cos\left(\frac{\pi (2N-k)(n+1/2)}{N}\right) = \sum_{n=0}^{N-1} h[n] \cos\left(\frac{\pi 2N(n+1/2)}{N} - \frac{\pi k(n+1/2)}{N}\right) \\ &= \sum_{n=0}^{N-1} h[n] \cos\left((2n+1)\pi - \frac{\pi k(n+1/2)}{N}\right) = - \sum_{n=0}^{N-1} h[n] \cos\left(\frac{\pi k(n+1/2)}{N}\right) = -\tilde{H}[k] \end{aligned}$$

In particular $\tilde{H}[N] = 0$

Inverse Type II DCT

The inverse transform is

$$\begin{aligned} h[n] &= \frac{1}{2N} \sum_{k=0}^{2N-1} H[k] \exp\left(-i \frac{2\pi kn}{2N}\right) \\ &= \frac{1}{N} \sum_{k=0}^{2N-1} \frac{1}{2} H[k] \exp\left(i \frac{\pi k}{2N}\right) \exp\left(-i \frac{2\pi k(n+1/2)}{2N}\right) \\ &= \frac{1}{N} \sum_{k=0}^{2N-1} \tilde{H}[k] \exp\left(-i \frac{2\pi k(n+1/2)}{2N}\right) \\ &= \frac{1}{N} \left[\sum_{k=0}^{N-1} \tilde{H}[k] \exp\left(-i \frac{2\pi k(n+1/2)}{2N}\right) + \sum_{k=N}^{2N-1} \tilde{H}[k] \exp\left(-i \frac{2\pi k(n+1/2)}{2N}\right) \right] \\ &= \frac{1}{N} \left[\sum_{k=0}^{N-1} \tilde{H}[k] \exp\left(-i \frac{2\pi k(n+1/2)}{2N}\right) + \sum_{k=N}^{2N-1} \tilde{H}[2N-k] \exp\left(i \frac{2\pi(2N-k)(n+1/2)}{2N}\right) \right] \end{aligned}$$

Change of notation of variable $2N - k \rightarrow k$ in the second term:

$$\begin{aligned} h[n] &= \frac{1}{N} \left[\sum_{k=0}^{N-1} \tilde{H}[k] \exp\left(-i \frac{2\pi k(n+1/2)}{2N}\right) + \sum_{k=1}^N \tilde{H}[k] \exp\left(i \frac{2\pi k(n+1/2)}{2N}\right) \right] \\ &= \frac{1}{N} \left[\tilde{H}[0] + \sum_{k=1}^{N-1} \tilde{H}[k] \exp\left(-i \frac{2\pi k(n+1/2)}{2N}\right) + \sum_{k=1}^{N-1} \tilde{H}[k] \exp\left(i \frac{2\pi k(n+1/2)}{2N}\right) \right] \\ &= \frac{1}{N} \left[\tilde{H}[0] + 2 \sum_{k=1}^{N-1} \tilde{H}[k] \cos\left(\frac{\pi k(n+1/2)}{N}\right) \right] \end{aligned}$$

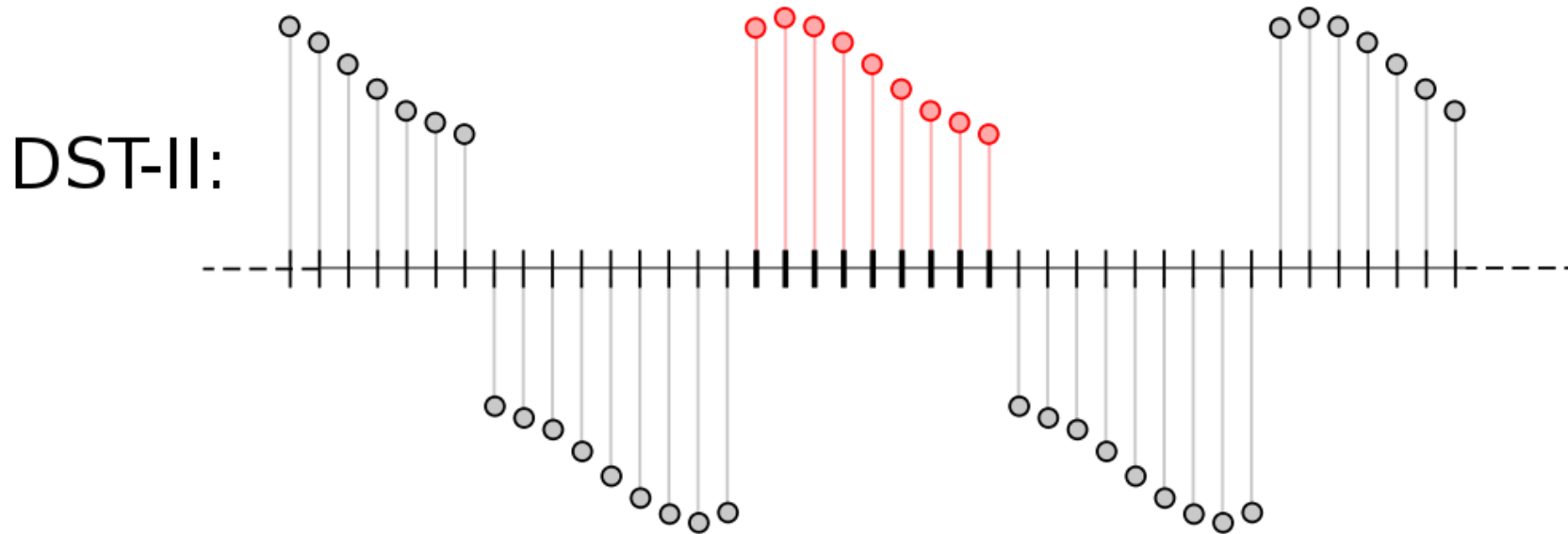
Type II vs Type I

- Type II DCT is more elegant and is the version that is most often used
- Notice that it transforms N inputs $h[n]$ to N outputs $\tilde{H}[k]$

Discrete Sine Transform (DST)

- Used much less often than cosine transform
- This is because it forces the function to be zero at either end of its range
- It is not true for most functions encountered in real-world applications, and so the sine transform has limited applicability

Type II DST



- $h[2N - n - 1] = -h[n]$
- Recall: One can choose any $2N$ frequencies k with $k \pmod{2N} = 0, 1, \dots, 2N - 1$
- In type II DST we choose $1, 2, \dots, 2N$ but keep using the index $k = 0, 1, \dots, 2N - 1$

Type II DST

$$\begin{aligned} H[k] &= \sum_{n=0}^{2N-1} h[n] \exp\left(i \frac{2\pi(k+1)n}{2N}\right) \\ &= \sum_{n=0}^{N-1} h[n] \exp\left(i \frac{2\pi(k+1)n}{2N}\right) + \sum_{n=N}^{2N-1} h[n] \exp\left(i \frac{2\pi(k+1)n}{2N}\right) \\ &= \sum_{n=0}^{N-1} h[n] \exp\left(i \frac{2\pi(k+1)n}{2N}\right) - \sum_{n=N}^{2N-1} h[2N-n-1] \exp\left(-i \frac{2\pi(k+1)(2N-n)}{2N}\right) \end{aligned}$$

Change of notation of variable $2N-n-1 \rightarrow n$ in the second term:

$$\begin{aligned} H[k] &= \sum_{n=0}^{N-1} h[n] \exp\left(i \frac{2\pi(k+1)n}{2N}\right) - \sum_{n=0}^{N-1} h[n] \exp\left(-i \frac{2\pi(k+1)(n+1)}{2N}\right) \\ &= \exp\left(-i \frac{\pi(k+1)}{2N}\right) \left[\sum_{n=0}^{N-1} h[n] \exp\left(i \frac{2\pi(k+1)(n+1/2)}{2N}\right) - \sum_{n=0}^{N-1} h[n] \exp\left(-i \frac{2\pi(k+1)(n+1/2)}{2N}\right) \right] \\ &= 2i \exp\left(-i \frac{\pi(k+1)}{2N}\right) \sum_{n=0}^{N-1} h[n] \sin\left(\frac{\pi(k+1)(n+1/2)}{N}\right) \end{aligned}$$

Type II DST

The prefactor is conventionally absorbed into the Fourier coefficients by defining

$$\tilde{H}[k] = \frac{1}{2i} H[k] \exp\left(i \frac{\pi(k+1)}{2N}\right) = \sum_{n=0}^{N-1} h[n] \sin\left(\frac{\pi(k+1)(n+1/2)}{N}\right)$$

which are real if $h[n]$ are real.

Also

$$\tilde{H}[2N-1] = 0$$

$$\begin{aligned} \tilde{H}[2N-k-2] &= \sum_{n=0}^{N-1} h[n] \sin\left(\frac{\pi(2N-k-1)(n+1/2)}{N}\right) = \sum_{n=0}^{N-1} h[n] \sin\left(\frac{\pi(2N)(n+1/2)}{N} - \frac{\pi(k+1)(n+1/2)}{N}\right) \\ &= \sum_{n=0}^{N-1} h[n] \sin\left((2n+1)\pi - \frac{\pi(k+1)(n+1/2)}{N}\right) = \sum_{n=0}^{N-1} h[n] \sin\left(\frac{\pi(k+1)(n+1/2)}{N}\right) = \tilde{H}[k] \end{aligned}$$

Inverse Type II DST

The inverse transform is

$$\begin{aligned} h[n] &= \frac{1}{2N} \sum_{k=0}^{2N-1} H[k] \exp\left(-i \frac{2\pi(k+1)n}{2N}\right) \\ &= \frac{i}{N} \sum_{k=0}^{2N-1} \frac{1}{2i} H[k] \exp\left(i \frac{\pi(k+1)}{2N}\right) \exp\left(-i \frac{2\pi(k+1)(n+1/2)}{2N}\right) \\ &= \frac{i}{N} \sum_{k=0}^{2N-1} \tilde{H}[k] \exp\left(-i \frac{2\pi(k+1)(n+1/2)}{2N}\right) \\ &= \frac{i}{N} \left[\sum_{k=0}^{N-1} \tilde{H}[k] \exp\left(-i \frac{2\pi(k+1)(n+1/2)}{2N}\right) + \sum_{k=N}^{2N-2} \tilde{H}[k] \exp\left(-i \frac{2\pi(k+1)(n+1/2)}{2N}\right) \right] \\ &= \frac{i}{N} \left[\sum_{k=0}^{N-1} \tilde{H}[k] \exp\left(-i \frac{2\pi(k+1)(n+1/2)}{2N}\right) + \sum_{k=N}^{2N-2} \tilde{H}[2N-k-2] \exp\left(-i \frac{2\pi(k+1)(n+1/2)}{2N}\right) \right] \end{aligned}$$

Inverse Type II DST

Change of notation of variable $2N - k - 2 \rightarrow k$ in the second term:

$$\begin{aligned} h[n] &= \frac{i}{N} \left[\sum_{k=0}^{N-1} \tilde{H}[k] \exp \left(-i \frac{2\pi(k+1)(n+1/2)}{2N} \right) + \sum_{k=0}^{N-2} \tilde{H}[k] \exp \left(-i \frac{2\pi(2N-k-1)(n+1/2)}{2N} \right) \right] \\ &= \frac{i}{N} \left[\sum_{k=0}^{N-1} \tilde{H}[k] \exp \left(-i \frac{2\pi(k+1)(n+1/2)}{2N} \right) - \sum_{k=0}^{N-2} \tilde{H}[k] \exp \left(i \frac{2\pi(k+1)(n+1/2)}{2N} \right) \right] \\ &= \frac{i}{N} \left[\tilde{H}[N-1] \exp \left(-i \frac{2\pi N(n+1/2)}{2N} \right) + \sum_{k=0}^{N-2} \tilde{H}[k] \exp \left(-i \frac{2\pi(k+1)(n+1/2)}{2N} \right) - \sum_{k=0}^{N-2} \tilde{H}[k] \exp \left(i \frac{2\pi(k+1)(n+1/2)}{2N} \right) \right] \\ &= \frac{i}{N} \left[\tilde{H}[N-1] \exp \left(-i\pi \left(n + \frac{1}{2} \right) \right) - 2i \sum_{k=0}^{N-2} \tilde{H}[k] \sin \left(\frac{\pi(k+1)(n+1/2)}{N} \right) \right] \\ &= \frac{1}{N} \left[\tilde{H}[N-1] \exp(-in\pi) + 2 \sum_{k=0}^{N-2} \tilde{H}[k] \sin \left(\frac{\pi(k+1)(n+1/2)}{N} \right) \right] \\ &= \frac{2}{N} \left[\frac{(-1)^n}{2} \tilde{H}[N-1] + \sum_{k=0}^{N-2} \tilde{H}[k] \sin \left(\frac{\pi(k+1)(n+1/2)}{N} \right) \right] \end{aligned}$$

Fast Fourier Transform (FFT)

Number of Operations in DFT

The basic definition of DFT $H[k] = \sum_{n=0}^{N-1} h[n] e^{2\pi i n k / N}$

gives rise to a matrix equation

$$\begin{pmatrix} H[0] \\ H[1] \\ \vdots \\ H[N-1] \end{pmatrix} = \begin{pmatrix} 1 & 1 & 1 & \dots & 1 \\ 1 & e^{2\pi i / N} & e^{4\pi i / N} & \dots & e^{2(N-1)\pi i / N} \\ 1 & e^{4\pi i / N} & e^{8\pi i / N} & \ddots & e^{4(N-1)\pi i / N} \\ \vdots & \vdots & \vdots & & \vdots \\ 1 & e^{2(N-1)\pi i / N} & e^{4(N-1)\pi i / N} & \dots & e^{(N-1)^2 \pi i / N} \end{pmatrix} \begin{pmatrix} h[0] \\ h[1] \\ \vdots \\ h[N-1] \end{pmatrix}$$

$\therefore$ The number of operation is $O(N^2)$

Fast Fourier Transform (FFT)

$$\boxed{N = 2^p} \rightarrow \text{Number of operations } \boxed{O(N \log N)}$$

For even N , DFT of length N can be rewritten as the sum of two DFT of length $N / 2$

$$\begin{aligned} H[k] &= \sum_{n=0}^{N-1} e^{2\pi i n k / N} h[n] \\ &= \sum_{n=0}^{N/2-1} e^{2\pi i (2n) k / N} h[2n] + \sum_{n=0}^{N/2-1} e^{2\pi i (2n+1) k / N} h[2n+1] \\ &= \sum_{n=0}^{N/2-1} e^{2\pi i n k / (N/2)} h[2n] + e^{2\pi i k / N} \sum_{n=0}^{N/2-1} e^{2\pi i n k / (N/2)} h[2n+1] \\ &= H^0[k] + e^{2\pi i k / N} H^1[k] = H^0[k] + W^k H^1[k], \quad (W \equiv e^{2\pi i / N}) \end{aligned}$$

The superscript $0 \rightarrow \text{even}$, $1 \rightarrow \text{odd}$

Fast Fourier Transform (FFT)

When k runs from 0 to $N-1$

$$H[k]: \text{Period } N \rightarrow H[0], H[1], \dots, H[N-1]$$

$$H^0[k], H^1[k]: \text{Period } N/2$$

$$\rightarrow \{H^0[0], H^0[1], \dots, H^0[N/2-1]\} \{H^1[0], H^1[1], \dots, H^1[N/2-1]\}$$

→ If the N numbers $H^0[k], H^1[k]$ are already known,

can generate the N numbers $H[k]$ by

$$H[k] = H^0[k] + W^k H^1[k]$$

Number of operation = N

Fast Fourier Transform (FFT)

$H^0[k], H^1[k]$ are themselves DFT with length $N/2$ which is still a power of 2

$\Rightarrow$ The same procedure can be carried out again

$\Rightarrow H^{00}[k], H^{01}[k], H^{10}[k], H^{11}[k]$: Period $N/4 \rightarrow N$ new numbers again

Each time, use the N numbers in a certain level to generate the N numbers in the next level

If $N = 2^p \Rightarrow$ terminate after p iterations

$\Rightarrow p$ levels

$\Rightarrow$ e.g., $H^{\overbrace{011001 \dots 101}^{p \text{ digits}}}[k] = h[n]$

$\therefore$ Number of operations $= pN = N \log N$

Example

$$N = 8$$

$$H[k] = \sum_{n=0}^7 W^{nk} h[n] = h[0] + W^k h[1] + W^{2k} h[2] + W^{3k} h[3] + W^{4k} h[4] + W^{5k} h[5] + W^{6k} h[6] + W^{7k} h[7]$$

Period 4, only need $k=0,1,2,3$. Total: 8 numbers

$$= \overbrace{(h[0] + W^{2k} h[2] + W^{4k} h[4] + W^{6k} h[6])}^{H^0[k]} + W^k \overbrace{(h[1] + W^{2k} h[3] + W^{4k} h[5] + W^{6k} h[7])}^{H^1[k]}$$

Period 2, only need $k=0,1$. Total: 8 numbers

$$= \overbrace{(h[0] + W^{4k} h[4])}^{H^{00}[k]} + W^{2k} \overbrace{(h[2] + W^{4k} h[6])}^{H^{01}[k]} + W^k [\overbrace{(h[1] + W^{4k} h[5])}^{H^{10}[k]} + W^{2k} \overbrace{(h[3] + W^{4k} h[7])}^{H^{11}[k]}]$$

Period 1, the same for all k . Total: 8 numbers

$$= \overbrace{(\hat{h}[0])}^{H^{000}[k]} + W^{4k} \overbrace{(\hat{h}[4])}^{H^{001}[k]} + W^{2k} [\overbrace{(\hat{h}[2])}^{H^{010}[k]} + W^{4k} \overbrace{(\hat{h}[6])}^{HH^{011}[k]}] + W^k [\overbrace{(\hat{h}[1])}^{H^{100}[k]} + W^{4k} \overbrace{(\hat{h}[5])}^{H^{101}[k]} + W^{2k} [\overbrace{(\hat{h}[3])}^{H^{110}[k]} + W^{4k} \overbrace{(\hat{h}[7])}^{H^{111}[k]}]]]$$

How to Identify Neighboring Elements

The digits in the superscript denote in which group (even or odd)

$h[n]$ goes in each level

If the integer n is represented by binary digits,

whether it's even or odd depends solely on its last digit

$0 \rightarrow \text{even}, 1 \rightarrow \text{odd}$

$\therefore h[n] \Leftrightarrow H^s \Rightarrow n$ is the bit-reversal of s

Algorithm

- One-point transform: Sort data $h[n]$ in bit-reversed order
- Two-point transform: Combine adjacent pairs
- Four-point transform: Combine adjacent pairs
-
- Etc.

Fast Sine Cosine Transforms and FFT

- FFT functions are common and usually you need not write your own code
- This is also true for fast Cosine and Sine transforms but if needed, you can write your own functions easily by using FFT